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Applying Counterfactual Explanations and Multivariate Forecasting to Medical Prediction Tasks

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Abstract

Forecasting patient outcomes is a critical task in healthcare, with applications ranging from treatment planning to risk assessment. While recent advances in time series forecasting have explored counterfactual reasoning, existing approaches typically modify past observations, limiting their practical utility in clinical settings where future-oriented interventions are more actionable. This thesis introduces a novel framework for counterfactual time series forecasting, which generates plausible future trajectories by adjusting exogenous variables within the forecast horizon to achieve desired clinical outcomes.

We incorporate multiple forecasting models, such as SARIMAX, OLS, GRU, and N-BEATSx, to learn the relationships between exogenous inputs and target variables, enabling interpretable and constrained counterfactual generation. Extensive evaluation across three healthcare applications—glucose level forecasting using OhioT1DM and SimGlucose datasets, and mortality prediction in HFpEF patients using MIMIC-III—demonstrates the effectiveness and realism of the proposed method. In particular, N-BEATS consistently achieves superior predictive performance and produces counterfactuals that adhere closely to physiological norms.

Our findings show that effective counterfactual interventions typically involve coordinated changes across multiple variables and that realistic, health-aligned adjustments can be achieved without deviating significantly from natural data distributions. Despite promising results, challenges such as class imbalance and the need for clinical validation remain. We outline future directions to improve clinical relevance through domain expert collaboration, richer patient context integration, and adherence-aware optimization.

This work establishes a foundation for developing transparent, actionable, and personalized forecasting tools in healthcare, capable of simulating “what-if” scenarios to support better clinical decision-making.

Contents

1	Introduction	1
1.1	Diabetes	3
1.2	Heart failure with preserved ejection fraction	3
1.3	Counterfactual Explanations	4
1.4	Main Contributions	4
1.5	Problem Formulation	5
2	Related Work	5
2.1	Related Forecasting Models	6
2.2	Related Work in Glucose Forecasting	7
2.3	Related Work in Mortality Prediction of Patients with HFpEF	8
3	Methodology	9
3.1	Algorithm	9
3.2	Data	13
3.3	Experiments	18
3.4	Evaluation Metrics	23
4	Results	24
4.1	Multivariate Forecasting	24
4.2	Counterfactuals	29
4.3	Evaluation Metrics	32
5	Discussion	37
5.1	Multivariate Forecasting	37
5.2	Counterfactuals	39
5.3	Limitations and Future Work	43
6	Conclusion	43
	References	49
A	Appendix	49
A.1	Detailed Classification Metrics for MIMIC Dataset	49

1 Introduction

The goal of this thesis is to gain insights into the application of multivariate forecasting and counterfactual explanations to medical prediction tasks.

The increasing availability of large and complex healthcare datasets has accelerated the adoption of machine learning (ML) and deep learning (DL) techniques for clinical decision support [ASFWHY⁺25, MPS19]. In particular, time series forecasting plays a vital role in monitoring patient states and predicting clinical outcomes based on continuously recorded data, such as vital signs and laboratory results [KCS⁺20, JPS⁺16]. However, despite the progress in predictive accuracy, many DL models remain black boxes, limiting their interpretability and acceptance in critical healthcare environments.

Counterfactual explanations (CEs) have emerged as a promising approach to enhance model interpretability by identifying minimal and actionable changes in input features that would lead to a more desirable prediction [VDH21, Gui24]. While CEs have been widely studied in static tabular data contexts, their extension to time series forecasting, especially in multivariate medical data, is still under explored. Existing methods for counterfactual time series mainly focus on altering historical data points, which is impractical in real-world scenarios where the past is fixed [WSMP24].

This thesis aims to address this knowledge gap by developing a novel counterfactual forecasting framework that proposes actionable changes to exogenous variables during the forecast horizon, enabling dynamic and interpretable decision-making in healthcare. We leverage and integrate classical statistical models and state-of-the-art deep learning architectures to learn the relationships between target variables and exogenous inputs, facilitating constrained forecasting that aligns with clinical goals.

The global population is both growing and ageing, which is resulting in a rise in chronic diseases. This is resulting in a demand for more efficient, personalised, and proactive healthcare solutions [HSL⁺23]. Previously, treatment was only provided once symptoms had appeared, and treatment plans were very general. Nowadays, this approach is no longer sufficient, as patients are in need of a more personalised approach to their care. Technological advancements in the field of healthcare have had a significant impact on healthcare providers and patients, with statistical methods being used to predict outcomes. This approach struggles with the complex nature of clinical, demographic and molecular factors that influence the disease progression, leading to Machine Learning (ML), Deep Learning (DL), Artificial Intelligence (AI), and Big Data Analytics becoming increasingly popular fields within the medical and health science domains [ASFWHY⁺25]. Recent healthcare has been characterised by an increased need for data-driven approaches, with the care process being driven by the flow of data between patients and doctors, and the sharing of decisions, instructions and information amongst care providers. The role that data and information play in decision making and provision of healthcare, has only increased with the growing digitisation of healthcare. This results in great amounts of data, which enables the implementation of advanced analytical methods, including ML and AI, to derive valuable and actionable insights. These insights are essential in supporting decision-making processes, ensuring high quality patient care, responding to real-time situations and ultimately reducing mortality. Especially ML becomes more and more relevant in healthcare applications, including predictive analysis, treatment optimisation, and patient monitoring [MPS19]. ML algorithms can potentially be used to improve diagnostic accuracy, as well as support early disease detection and prediction. Other applications

include, analysing medical imaging data, such as X-rays and MRIs, to detect signs of cancer or neurological disorders, which allows for early diagnosis, as well as a personalised treatment plan for each patient. These uses show the potential ML has for both research and clinical trials, and support the improvement of healthcare overall [RNZ17], [JHS⁺22].

Time series analysis uses an ordered sequence of data points recorded over time, usually at regular intervals, to understand patterns, trends, and relationships within the data over time. This time series data also allows for forecasting or predicting future values, based on historical observations. Time series forecasting plays a crucial role in a number of different applications. Some applications include its usage in finance, for predicting stock prices and market trends [MSG14], in meteorology for weather and climate forecasting [KS20], in transportation, for effective traffic flow forecasting [LBF13] and especially recently in healthcare. In healthcare, time series forecasting is increasingly relevant, as patient data such as heart rate, blood pressure, glucose levels, and laboratory results are being collected continuously or in regular intervals. Precise forecasting of these medical time series can allow for early detection of negative outcomes, support the personalisation of treatment plans, and optimise healthcare management overall [KCS⁺20], [JPS⁺16].

[**TODO:** Extend, add citations] Generally, time series forecasting consists of finding temporal dependencies and trends in the data. Depending on the nature of the time series, such as for example linear or non-linear, stationary or non-stationary, and univariate or multivariate, requires different forecasting methods. These different forecasting methods can be categorised into three subgroups, statistical models, such as ARIMA and ETS, ML models, such as Linear Regression and SVRs, and DL models, such as RNNs, LSTMs, GRUs as well as transformer-based models Autoformer or DLinear.

Especially the latter models have shown promising results in recent years, achieving very accurate predictions. Despite these recent developments, many deep-learning based forecasting systems are considered "black-box"-models, as it is challenging to interpret and understand both the modelling process and the forecasting outcome. This is particularly problematic in healthcare applications, where especially interpretability is very important, as clinicians need to understand the reasoning behind the predictions to make informed decisions. One approach to tackle this problem is working with counterfactual explanations, which aim to identify small changes in the input variables needed to change a model's forecast to a more desirable outcome.

In healthcare, time series forecasting has been applied to a number of different tasks, especially the analysis of electronic health records (EHRs). These contain information about a patient's health over time, allowing predictions about the disease progression, treatment efficacy or patient mortality. For example, forecasting glucose levels in diabetic patients or predicting heart failure with preserved ejection fraction (HFpEF) can aid early interventions and optimise treatment plans.

Traditional time series forecasting focuses mainly on predicting future values given historical observations, but modifying past values is not feasible in real-world settings, especially in healthcare applications. Instead, a more practical approach is to explore how changing exogenous variables during the forecast horizon could lead to a desired outcome. This approach allows continuous monitoring of patients, making it possible to dynamically adjust treatment plans to lead towards more optimal results.

1.1 Diabetes

Diabetes is one of the most prevalent chronic diseases in the world, with it being a leading cause of death and disability. According to the World Health Organisation (WHO), around 830 million people worldwide suffer from diabetes, and it is the direct cause of 1.5 million deaths a year. Nowadays, it affects around 14% of adults, this number has doubled since 1990, making it a major public health problem. Diabetes is characterised by elevated levels of blood glucose, which over time seriously damages the heart, blood vessels, eyes, kidneys and nerves. Type 2 diabetes is much more common and occurs when the body either becomes resistant to insulin or does not make enough insulin. It usually has a later onset and its development can be attributed to factors such as being overweight, not getting enough exercise and genetic. Type 1 Diabetes is a chronic condition, where the pancreas only produces little to no insulin by itself. It is caused by the autoimmune destruction of pancreatic β -cells and affects 5-10% of the diabetes patients [Dia], [EMA+24a]. For patients with conditions such as type 1 diabetes mellitus (T1DM), closely tracking their glucose levels is a necessity. To reduce the risk of complications such as hyperglycaemia or hypoglycaemia, these patients rely on continuous glucose monitoring (CGM) devices and automated insulin delivery. Using machine learning (ML) to model CGM, can help to gain a better understanding of predicting abnormal glucose events and help with insulin dosage planning. By also incorporating variables such as insulin intake, carbohydrate consumption, and physical activity, a predictive model can allow timely interventions through the generation of actionable recommendations for patients or healthcare providers. This allows for more dynamic treatment based on these forecasted expected glucose trends, which can reduce the long-term risk of diabetes-related complications [EMA+24b].

1.2 Heart failure with preserved ejection fraction

Heart failure with preserved ejection fraction (HFpEF) is a prevalent and severe cardiovascular condition [BP10]. Heart failure (HF) is classified based on the left ventricular ejection fraction (LVEF) and can be split into three subcategories. Heart failure with reduced ejection fraction (HFrEF) with $LVEF \leq 40\%$, heart failure with mildly reduced ejection fraction (HFmrEF) with $LVEF = 41-49\%$, and heart failure with preserved ejection fraction (HFpEF) with $LVEF \geq 50\%$ [MMA+21]. Studies have shown that the mortality within one year is around 29%, [OHH+06], [SBA+24], with increased mortality for patients with previous heart failure hospitalisations and other comorbidities [MGL+20]. This makes HFpEF a very serious condition, where early diagnosis is key. So another possible application of medical time series forecasting could be to try identifying early warning signs of heart failure or more specific HFpEF to suggest either lifestyle or treatment changes. Using the vital signs, as for example the heart rate and blood pressure, of a patient as well as other factors like gender and possible comorbidities, allows specific and personal monitoring of disease progression. This way, early warning signs of worsening heart condition can be identified and personalised modifications to lifestyle or medication can be suggested. Such proactive monitoring can help reduce hospital readmissions and improve patient outcomes.

1.3 Counterfactual Explanations

Counterfactual explanations (CE) are an emerging technique with the potential to improve the interpretability and explainability of ML models. The objective is to identify minimal changes to the input features that would result in a different, and typically more desirable, outcome. More specifically, they provide information such as if an input data point would be X' instead of X , then the trained ML models prediction would be Y' instead of Y , assuming outcome Y' would be more favourable [VDH21]. So counterfactuals are a type of explanation method in ML, that might help users to understand the model predictions and decisions better. This is supported by counterfactual explanations being actionable, since they suggest specific changes to alter the outcome, and intuitive for humans, since they align with human reasoning about cause and effect. Often CEs are also model-agnostic, meaning they can be applied to a variety of black-box models [Gui24]. A more specific example application of counterfactuals would be a suggestion that slightly reducing a patients systolic blood pressure could have led to a prediction of a lower cardiovascular risk. So these kind of insight can not only help clinicians understand a models decisions better, but also provide help for intervention and treatment planning.

1.4 Main Contributions

[**TODO: Check, rewrite?**] Time series forecasting can be quite useful in healthcare by predicting how well a treatment works or predicting the risk of complications, relapse or mortality. Recent studies, such as COMET [WSMP24], have been focusing on counterfactuals in time series forecasting, but these methods work by altering historical observations. Our work aims to bridge this gap by developing a counterfactual forecasting mechanism that identifies optimal changes in exogenous variables during the forecast horizon to achieve desirable outcomes. More specifically, we propose a method that learns the relationship between the forecasted targets and exogenous variables, which leads to a more effective and interpretable decision-making in healthcare.

The main contributions of this thesis can be summarised as follows:

- We propose a new model for counterfactual time series forecasting to achieve a desired constrained forecast by modifying exogenous variables within the forecast horizon.
- We incorporate existing forecasting models, such as SARIMAX, OLS, GRU and NBEATSx, for learning the relationship between exogenous variables and a target variable to ensure actionable and interpretable predictions.
- We evaluate the models on two applications in healthcare, specifically for glucose level prediction and HFpEF management, demonstrating the practical utility of our approach.

1.5 Problem Formulation

1) Let $\mathbf{X} := (\mathbf{x}_i)_{i \in \{1, \dots, n\}}$ be a time series of length n (*back horizon*), with each $x_i \in \mathbb{R}^{m+1}$ composed of the target variable $y_i \in \mathbb{R}$ and the exogenous variables

$$z_i = \begin{pmatrix} z_{1,i} \\ \vdots \\ z_{m,i} \end{pmatrix} \in \mathbb{R}^m.$$

Then $\mathbf{X}$ can be denoted as the combined matrix of the target vector $\mathbf{y} \in \mathbb{R}^{1 \times n}$ and the exogenous matrix $\mathbf{Z} \in \mathbb{R}^{m \times n}$:

$$\mathbf{X} = \begin{pmatrix} \mathbf{y} \\ \mathbf{Z} \end{pmatrix} := \begin{pmatrix} y_1 & \dots & y_n \\ z_{1,1} & \dots & z_{1,n} \\ \vdots & \ddots & \vdots \\ z_{m,1} & \dots & z_{m,n} \end{pmatrix}.$$

The relationship between $\mathbf{y}$ and $\mathbf{Z}$ can be described by the function:

$$r : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{1 \times n}$$

which calculates the target vector from the exogenous matrix:

$$r(\mathbf{Z}) = \mathbf{y}.$$

2) Given a time series forecasting model f that predicts the next t values (*forecasting horizon*) of $\mathbf{X}$, we define the forecast as:

$$f(\mathbf{X}) = \mathbf{X}' := (x_{n+i})_{i \in \{1, \dots, t\}}.$$

Additionally, consider a lower and upper bound set of constraints for each time step in the forecasting horizon, denoted as:

$$\boldsymbol{\alpha} = (\alpha_{n+i})_{i \in \{1, \dots, t\}}, \boldsymbol{\beta} = (\beta_{n+i})_{i \in \{1, \dots, t\}}.$$

The objective is to generate a counterfactual time series sample $\mathbf{Z}^*$, such that $\mathbf{y}^* = r(\mathbf{Z}^*)$ satisfies the given bounds:

$$\alpha_i \leq y_i^* \leq \beta_i, \forall y_i^* \in \mathbf{y}^*, i \in \{n+1, \dots, n+t\}.$$

Summarized research objective: Given a target vector $\mathbf{y}$ affected by the exogenous matrix $\mathbf{Z}$, a forecast horizon t , the original forecasted vector $\mathbf{y}'$ and the original forecasted exogenous matrix $\mathbf{Z}'$, the goal is to modify $\mathbf{Z}'$ to $\mathbf{Z}^*$ such that the corresponding target vector $\mathbf{y}^*$ is within constraints $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$.

2 Related Work

[TODO: check, add citations, rewrite]

2.1 Related Forecasting Models

Recent research has explored various deep learning (DL) models for time series forecasting, including recurrent neural network (RNN)-based models such as gated recurrent units (GRU) and long short-term memory (LSTM), as well as attention-based architectures like transformers. Transformer-based models, including Autoformer and Informer, have demonstrated strong performance in both univariate and multivariate forecasting tasks by capturing long-range dependencies more effectively than traditional RNN approaches. In the clinical domain, deep learning models have been applied extensively to glucose forecasting. For instance, Deep Multi-Output Forecasting [FAJ⁺18] introduced a multi-step forecasting framework that explicitly models the distribution of future glucose values over a prediction horizon using a multi-output deep architecture. Similarly, WaveNet has been adapted for glucose forecasting by leveraging dilated convolutional neural networks (CNNs) to model long-term dependencies [ZLH⁺18]. In addition, transfer learning techniques have been employed to enhance predictive performance by fine-tuning pre-trained models on patient-specific data while incorporating exogenous covariates such as insulin dosage and carbohydrate intake [MB20].

Beyond predictive performance, explainability remains a critical challenge in deep learning-based forecasting models. Traditional statistical models, such as ARIMAX and VARIMAX, are able to quantify relationships between exogenous factors and the target variable, but their forecasting accuracy is often outperformed by DL approaches. Recent research has focused on integrating explainability into forecasting models to combine the strengths of both interpretability and predictive performance. For example, NBEATSx extends the NBEATS framework by incorporating exogenous variables into its deep architecture, enabling a more structured decomposition of trend and seasonality. However, its interpretability remains static and does not fully capture the dynamic nature of forecasting outcomes.

To address the need for explainability, counterfactual explanations have gained traction in time series analysis. Initial efforts focused on time series classification, where counterfactuals were generated through instance-based modifications and gradient-based perturbations [AALC20]. This was done by introducing a framework for generating counterfactual explanations for multivariate time series classification, identifying minimal input modifications needed to alter the model’s decision, providing interpretability for high-dimensional time series models.

More recently, counterfactual explanations have been extended to time series forecasting. ForecastCF [WMSP23] proposed a deep learning-based method for generating counterfactuals in time series forecasting by identifying minimal input changes required to achieve desired prediction outcomes. Building on this, COMET [WSMP24] extended counterfactual explanations to multivariate time series forecasting, focusing on modifying exogenous variables (e.g., insulin, carbohydrates, and exercise) to generate actionable recommendations for glucose management.

In addition to counterfactual forecasting, diagnostic tools like TimeTuner [HSYZ23] have been developed to analyse how time representations influence forecasting models. By employing counterfactual explanations, TimeTuner enables the evaluation of multivariate time series representations and their impact on model predictions.

Despite these advances, counterfactual explanations for multivariate time series analysis remain an emerging research area. While existing methods demonstrate the feasibility

of generating counterfactuals for univariate forecasts, their generalisation to multivariate forecasting and real-world clinical applications remains limited. This work aims to extend on these existing methods by integrating counterfactual reasoning with multivariate forecasting models, focusing on modifying exogenous variables within the prediction horizon to provide actionable and interpretable interventions.

2.2 Related Work in Glucose Forecasting

Forecasting physiological indicators such as blood glucose levels, is crucial for managing diabetes. Time series forecasting and model explainability are becoming increasingly important in this field of medical prediction tasks, as in many others. Recent contributions to this area of research highlight the growing use of multivariate machine learning models to improve predictive accuracy and personalisation. This shows the growing need for interpretable and actionable insights, which aligns closely with the goals of this thesis.

Recent research highlights the growing development of multivariate and deep learning-based models for predicting glucose levels. For example, Kalita and Mirza [KM25] proposed a model that combines multi-head attention layers with neural basis expansion networks, capturing complex temporal and cross-feature dependencies in glucose data. Similarly, Benaïda et al. [BAI25] demonstrated the effectiveness of deep learning architectures for both single- and multi-step glucose forecasting, emphasising the importance of longer-term prediction capabilities in real-world applications. These multivariate models are consistent with the focus of this thesis on leveraging multiple signals, such as past glucose levels, physiological parameters, and contextual variables, for accurate forecasting. Personalisation has emerged as a key factor in clinical forecasting settings, as patient diversity affects model performance. Shen and Kleinberg [SK25a] addressed this issue by using incrementally retrained LSTM networks that adapt to each individual’s glucose dynamics. This improves performance, even when the CGM data is limited. Lara-Abelenda et al. [LCP⁺25] introduced large language models (LLMs) to model personal glucose trends, highlighting the capacity of foundation models to generalise across individuals while retaining patient-specific nuances. These methods emphasise the importance of adaptive and context-aware forecasting.

Several works have also incorporated physiological signals beyond glucose levels to support multivariate forecasting. For example, Giancotti et al. [GBV⁺24] explored the utility of heart rate as a predictor of forecasting glucose levels in patients with type 1 diabetes, which demonstrates that multimodal data can significantly enhance predictive accuracy. Similarly, Rodríguez-Rodríguez et al. [RCR23] utilised data, such as physical activity and diet logs, to enable more holistic and personalised glycaemic forecasting.

Interpretability remains a major challenge for deep learning-based forecasting models, especially in critical fields such as medicine. In response to this, Sun and Kosmas [SK25b] combined Bayesian forecasting with expert medical knowledge to model CGM values in type 2 diabetes patients. Their framework improves both uncertainty quantification and clinician interpretability, which is an essential consideration in healthcare AI. The need for model transparency directly motivates the use of counterfactual explanations to improve the explainability and actionability of predictive models in medical applications. While many current studies emphasise predictive accuracy, fewer address how predictions can be explained and acted upon by clinicians or patients.

Taken together, these studies reflect a shift towards data-driven, multivariate, and person-

alised models for medical forecasting. However, there remains a clear gap in integrating these powerful models with robust, interpretable explanations. This thesis aims to bridge this gap by combining multivariate forecasting approaches with counterfactual reasoning, to provide accurate predictions and actionable, understandable explanations, which are essential components for supporting medical decision-making and patient self-management.

2.3 Related Work in Mortality Prediction of Patients with HFpEF

Heart Failure with Preserved Ejection Fraction (HFpEF) is a complex and heterogeneous condition, characterised by diagnostic and prognostic uncertainty. This makes it a compelling use case for machine learning (ML) in clinical decision support. Recent research has applied various ML techniques to improve diagnosis, predict outcomes such as hospitalisation and mortality, and guide individualised management strategies. These efforts emphasise the increasing relevance of multivariate forecasting and the growing demand for explainable models, which are central to the objectives of this thesis.

A significant amount of research has focused on prognostic modelling using structured clinical data. For example, Hu et al. [HMH⁺25] developed and validated a machine learning model to predict the risk of readmission within one year for HFpEF patients, demonstrating the utility of routinely collected electronic health records (EHRs) in anticipating adverse outcomes. Similarly, McDowell et al. [MKT⁺24] constructed models for predicting both mortality and morbidity in HFpEF patients, showing that complex risk factors, including comorbidities and laboratory values, can be effectively integrated into predictive models. These studies emphasise the importance of leveraging multivariate data sources to forecast long-term patient outcomes.

Short-term outcome prediction has also been explored, particularly in the context of the early identification of high-risk patients. The study "Predicting 30-Day and 1-Year Mortality in HFpEF" [SBA⁺24] used machine learning to predict short-term mortality, which is essential for planning acute care. Other models have focused on hospitalisation prediction, using historical patient trajectories to anticipate future events. These forecasting tasks not only require accurate time series modelling but also benefit from interpretability to inform clinical decisions.

The diagnosis of HFpEF remains a challenging area due to its symptomatic overlap with other heart failure subtypes. Kavas et al. [KBKB23] developed an ML-based decision support system using photoplethysmography (PPG) signals to differentiate between HFpEF and HFrEF (Heart Failure with reserved Ejection Fracion), demonstrating the potential of non-invasive, sensor-based diagnostics. Liao and Hung [LH⁺24] further extended this approach by incorporating data from a wearable patch device to enhance diagnostic precision. These works highlight the growing role of physiological signal data in heart failure classification, which directly supports multivariate modelling approaches by introducing continuous and high-frequency signals into prediction tasks.

Genomic and molecular data have also been used to support precision medicine approaches in HFpEF. Zhou et al. [ZGW⁺21] utilised gene expression profiles to build ML models capable of risk stratification in HFpEF patients, adding a layer of biological interpretability to purely clinical models. Although these models are powerful, they are often perceived as "black boxes," emphasising the need for explainability techniques such as counterfactual explanations to bridge the gap between model prediction and clinical insight.

Across these studies, however, the challenge of model transparency and interpretability remains largely unaddressed. Most existing models prioritise predictive performance without offering sufficient explanations for individual predictions, which is a critical issue in medical contexts where understanding why a prediction was made is often as important as the prediction itself. This thesis aims to bridge this gap by integrating counterfactual reasoning into multivariate forecasting models, offering clinicians not just a forecast, but a clear explanation of the factors driving the prediction and the minimal changes that could alter an adverse outcome.

In summary, the current research in HFpEF prediction demonstrates the power of machine learning to handle complex, multivariate data across diagnostic and prognostic applications. However, a lack of interpretability limits clinical adoption. This thesis tries to contribute to the field by combining accurate time series forecasting with interpretable, counterfactual explanations, thereby supporting more transparent and actionable decision-making in heart failure care.

3 Methodology

3.1 Algorithm

[TODO: update, check] The proposed algorithm is designed to generate counterfactual exogenous inputs that lead to a desired change in the forecasted target variable within a multivariate time series context. It operates by integrating a differentiable forecasting model and a counterfactual optimization process constrained by both feasibility and historical plausibility. The algorithm takes multivariate time series data as input, which is split into a target variable t (for example blood glucose level) and associated exogenous variables E (for example insulin dosage, carbohydrate intake, activity level). Several hyperparameters are defined: the learning rate η , target bounds (α, β) , a clipping range (ρ, ϕ) , and the maximum number of optimization iterations `max_iter`. Additionally, differentiable models for forecasting $f(\cdot)$ and counterfactual generation $c(\cdot)$ are provided, where f predicts the target and c estimates the outcome, given perturbed inputs. The goal is to identify a perturbed exogenous input E' that results in a forecasted target t' within the desired bounds. The algorithm consists of different stages:

- **Multivariate Forecasting:** The forecasting model f is applied to the input sequence to obtain predicted targets $\hat{t}^*$ and exogenous outputs $\hat{E}^*$. These serve as a reference for evaluating the effect of the perturbations.
- **Sample and Bound Selection:** A representative subset $\mathcal{S}$ of the test data is selected to compute statistical bounds (α, β) on the desired target outcome. These bounds guide the optimization process by defining what is considered a successful counterfactual prediction
- **Loss Initialization:** A composite loss function L is initialized. It incorporates forecasting error, a regularization term to penalize unrealistic perturbations, an activity temporal constraint and a historical similarity constraint.
- **Iterative Counterfactual Optimization:** Counterfactuals are generated through gradient-based optimization, with the Adam optimizer. In each iteration the loss is back

propagated with respect to $\hat{E}^*$, and updates are applied. Then a clipping function ensures that the perturbed values stay within predefined limits (ρ, ϕ) and a historical similarity constraint $\mathcal{G}$ encourages solutions to be close to real historical values, to enhance realism and interpretability.

- **Stopping criteria:** The loop terminates once the forecasted target falls within the desired bounds or when the maximum number of iterations is reached.
- **Return Output:** The final optimized exogenous input E' and its associated target t' are returned as the counterfactual explanation.

This algorithmic approach provides an adjustable and interpretable method for generating actionable interventions in time series forecasting tasks. It ensures that the generated counterfactuals are both effective, as they achieve the target, and feasible, as they are within realistic operational constraints, making it a suitable method for the delicate field of healthcare. Algorithm 1 shows the pseudocode for the described method.

Algorithm 1 Counterfactual Hybrid Forecasting

```

1: Input: Time series data: target  $t$ , exogenous variables  $E$ , learning rate  $\eta$ , desired
   bounds  $(\alpha, \beta)$ , clipping range  $(\rho, \phi)$ , max iterations max_iter, differentiable forecaster
    $f(\cdot)$  differentiable counterfactual generator  $c(\cdot)$ ,  $w$ ,  $\mathcal{G}$ 
2: Output: Counterfactual  $E'$  with desired outcome  $t'$ 
3:  $(t^*, E^*) \leftarrow (t, E)$ 
4:  $(\hat{t}^*, \hat{E}^*) \leftarrow f(t^*, E^*)$ 
5:  $\mathcal{S} \leftarrow \text{SelectTestSamples}$ 
6:  $(\alpha, \beta) \leftarrow \text{GenerateBounds}(\mathcal{S})$ 
7:  $C \leftarrow \text{ActivityTemporalConstraint}$ 
8:  $loss \leftarrow L((t^*, E^*), \alpha, \beta, (t, E), C)$ 
9: time  $\leftarrow 0$ 
10: while  $(\hat{t}^* > \beta \text{ or } \hat{t}^* < \alpha) \wedge (\text{time} < \text{max\_iter})$  do
11:    $\hat{E}^* \leftarrow \text{AdamOptimize}(\hat{E}^*, loss, \eta)$ 
12:    $\hat{E}^* \leftarrow \text{Clip}(\hat{E}^*, \rho, \phi)$ 
13:    $\hat{t}^* \leftarrow c(\hat{E}^*)$ 
14:    $C \leftarrow \text{HistValueConstraint}(\hat{E}^*, \mathcal{G})$ 
15:    $loss \leftarrow L(\hat{E}^*, w, \alpha, \beta, X^*, C)$ 
16:   time  $\leftarrow \text{time} + 1$ 
17: end while
18:  $(t', E') \leftarrow (\hat{t}^*, \hat{E}^*)$ 
19: return  $(t', E')$ 

```

3.1.1 Desired Bound Generation

To encourage realistic and safe counterfactual predictions, we created soft, personalized bounds for future glucose values. These bounds help guide the model’s outputs toward safe glucose ranges, adjusting based on each patient’s current state and allowing enough time for the changes to occur in a physiologically reasonable way. For the HFpEF problem, the target is mortality, and therefore the target is binary, either the patient survives or

the patient does not. Because of that, no bounds were generated for that problem, with survival being the desirable parameter in the gradient perturbation.

Let t_{n+1} denote the first predicted target value in the forecast horizon (i.e., the current forecasted level), and let t^* be the desired glucose target we aim to reach after S time steps. We construct a smooth transition function $b(time)$, based on a polynomial of order p , to define a target trend from t_{n+1} to t^* :

$$b(time) = t_{n+1} + (t^* - t_{n+1}) \cdot \left(\frac{t}{S}\right)^p, \quad t = 0, 1, \dots, t-1$$

This equation produces a smooth curve that starts at the current value and gradually transitions to the target. Once the target is reached at $time = S$, the curve remains flat at $time^*$ for the remaining horizon.

To account for variability and uncertainty in glucose behavior, we add a margin around this curve using the standard deviation σ of the input time series. This results in an upper and lower bound as follows:

$$\alpha_{n+time} = b(time) - \lambda \cdot \sigma, \quad \beta_{n+time} = b(time) + \lambda \cdot \sigma$$

Here, $\lambda \in [0, 1]$ is a tunable hyperparameter (usually $\lambda = 0.5$) that controls how wide the bounds are. All calculations are performed in scaled (normalized) space using the patient-specific scalers from training.

3.1.2 Constraints

[TODO: check, rewrite, improve] To ensure that the generated counterfactual trajectories are not only effective in altering predicted outcomes but also clinically feasible and temporally realistic, we introduce a set of constraint mechanisms directly into the optimization process. These are the clipping constraint, the historical value constraint, and the activity temporal constraint. Each plays a distinct role in maintaining interpretability, trustworthiness, and clinical plausibility of the counterfactuals.

These constraints are applied iteratively during optimization, guiding the perturbation of exogenous variables such that the resulting trajectories adhere to domain-specific boundaries and realistic temporal patterns.

3.1.2.1 Clipping Constraint

The clipping constraint ensures that the perturbed exogenous variables $\hat{E}^*$ remain within a realistic value range. These bounds (ρ, ϕ) are selected based on physical limits, clinical safety, or operational feasibility (e.g., insulin dosages must be non-negative and within a patient-safe maximum range).

After each optimization step, every element in $\hat{E}^*$ is clipped such that:

$$\hat{E}_*^{i,j,k} = \min(\phi_k, \max(\rho_k, \hat{E}_*^{i,j,k}))$$

for each sample i , time step j , and feature k . This ensures that the generated counterfactuals remain within a trusted domain of possible values.

3.1.2.2 Historical Value Constraint

The historical value constraint is introduced to promote plausibility by encouraging counterfactual inputs that are statistically similar to observed historical inputs. This constraint acts as a regularization mechanism, penalizing counterfactuals that deviate too far from past observed patterns. It leverages a historical dataset $\mathcal{G}$ to compute a similarity or distance metric, often based on Euclidean or Mahalanobis distance.

Formally, a constraint term C is computed based on the deviation of $\hat{E}^*$ from the nearest historical samples:

$$C(\hat{E}^*) = \lambda \cdot \min_{E \in \mathcal{G}} \|\hat{E}^* - E\|_2$$

where λ is a hyperparameter that controls the strength of this constraint. The closer $\hat{E}^*$ is to a real historical sample, the lower the penalty.

This mechanism ensures that the generated exogenous sequences are not only valid within a static range but also dynamically aligned with real-world behavior.

3.1.2.3 Activity Temporal Constraint

The activity temporal constraint imposes structure on *when* changes to features are allowed during the time series. This is useful in scenarios where only certain time steps are clinically actionable or when domain knowledge suggests that interventions (e.g., medication, exercise, meals) are expected at specific moments.

This constraint is encoded using a binary vector $C \in \{0, 1\}^T$, where each element C_j corresponds to a time step j . A value of $C_j = 0$ allows counterfactual perturbations at that time, while $C_j = 1$ discourages them.

We define the constraint as:

$$C_j = \begin{cases} 0 & \text{if } \hat{E}^j > \gamma \\ 1 & \text{otherwise} \end{cases}$$

where γ is a user-defined threshold (e.g., $\gamma = 0$) that determines whether the feature value at time j indicates meaningful activity. This ensures changes happen primarily at relevant time steps, avoiding unrealistic shifts in periods where intervention is clinically unlikely or inappropriate.

The corresponding activity penalty is computed as:

$$C_{\text{act}}(\hat{E}^*, E) = \sum_j C_j \cdot \left| \frac{\hat{E}^j - E^j}{E^j + \epsilon} \right|$$

where ϵ is a small constant to prevent division by zero. This encourages local changes to occur only when and where they are contextually valid.

3.1.2.4 Combined Loss Function

The final optimization objective integrates all the above constraints with the primary forecasting goal. The loss function is composed of:

- **Forecast loss:** Penalizes predictions $\hat{t}^*$ that fall outside the desired outcome bounds (α, β) .

- **Historical deviation penalty:** Encourages similarity to real historical trajectories.
- **Activity temporal penalty:** Controls when temporal changes can occur.

The total loss is defined as:

$$L = \text{ForecastLoss}(\hat{t}^*, \alpha, \beta) + \lambda_{\text{hist}} \cdot C_{\text{hist}}(\hat{E}^*) + \lambda_{\text{act}} \cdot C_{\text{act}}(\hat{E}^*, E)$$

Here, λ_{hist} and λ_{act} are hyperparameters that tune the importance of each constraint. The clipping constraint is applied through projection and is not explicitly included in the loss function.

These three constraints work together to guide the counterfactual generation process as the clipping constraint ensures that variable values remain in a trusted clinical range, the historical value constraint keeps the counterfactuals close to real, observed trajectories, and the activity temporal constraint encourages changes to occur at appropriate, clinically meaningful time steps. By embedding these constraints into the optimization loop, the generated counterfactuals are not only effective but also interpretable, trustworthy, and aligned with realistic clinical dynamics.

3.2 Data

While the proposed model has broad applicability across domains beyond healthcare, this study focuses exclusively on medical use cases. In particular, we investigate the models usefulness in optimizing treatment plans for two conditions: type 1 diabetes and heart failure with preserved ejection fraction (HFpEF). The datasets used in these experiments contain physiological measurements and treatment-related variables, allowing for personalized forecasts and counterfactual intervention generation. To prepare the data for modelling, several preprocessing steps have been applied. First, the data is split into training, validation and test sets and normalized using min-max scaling to ensure stable and consistent model training. The data is then separated into target variables and exogenous inputs. Then, a sequence generator is employed to segment the time series into overlapping windows comprising a back horizon (historical input) and a prediction horizon (target output). This is done both for the target variable alone and for sequences that include both the target and exogenous features. Any sequences containing missing values are discarded to ensure data integrity.

3.2.1 SimGlucose

The SimGlucose dataset is generated using the FDA-approved UVA/PADOVA type 1 diabetes simulator [Xie18], a Python-based tool that models the physiological responses of individuals with type 1 diabetes. The simulator includes 30 virtual patients, comprising of 10 adults, 10 adolescents, and 10 children and produces continuous glucose monitoring (CGM) measurements along with insulin dosages and carbohydrate intake events. The dataset is generated with a predefined CGM sampling frequency and insulin pump settings based on the algorithm developed in [DMML⁺14]. For this study, simulated data was generated for ten adult patients over a one-week period. The blood glucose (BG) levels serve as the primary target variable, while the insulin dosage and carbohydrate intake are used as exogenous variables influencing the glucose levels. After generation, the data

undergoes preprocessing steps as described above to prepare it for the forecasting and counterfactual generation. An example of the generated data is shown in Figure 1. In this example, BG denotes blood glucose levels, CHO indicates carbohydrate intake, and Insulin reflects the administered dosage. The risk index illustrates periods of hyperglycaemic or hypoglycaemic risk. The green band in the BG trace highlights the target glucose range (70–180 mg/dL), while red regions denote values that fall outside this range, corresponding to hypo- or hyperglycaemia.

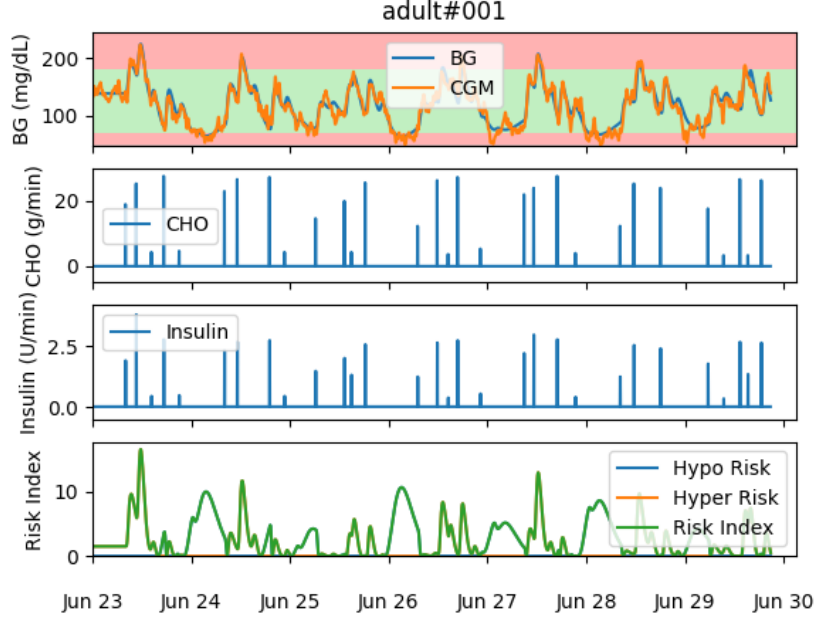


Figure 1: Simulation of an adult patient over the span of one week.

3.2.2 OhioT1DM:

The OhioT1DM dataset [MB20] contains real-world glucose monitoring data collected from 12 individuals with type 1 diabetes over an eight-week period by Ohio University. Following prior research [CHN⁺21], [WSMP24], we extracted the most clinically relevant features for forecasting: continuous glucose monitoring (CGM) measurements, basal insulin, bolus insulin, carbohydrate intake, and physical activity. Compared to the SimGlucose dataset, OhioT1DM includes a more varied set of exogenous variables, particularly basal and bolus insulin administration, dietary intake, and physical activity data. As a real-world dataset, it presents additional challenges such as missing values and irregular sampling intervals. These are addressed using interpolation and resampling techniques to ensure consistency in the input sequences. The inclusion of diverse exogenous variables enables the development of more nuanced counterfactual interventions and supports improved forecasting performance. Figure 2 illustrates a 24-hour time window for one patient, showing the temporal relationship between glucose levels, insulin administration, and carbohydrate consumption. The blue dotted line represents CGM-based blood glucose levels, while the black line represents the basal insulin, reflecting its slow-acting,

sustained delivery throughout the day. Orange dots indicate the amount of bolus insulin, while small blue boxes indicate meals. Spikes in blood glucose often correspond to meal times (carbohydrate intake), followed by bolus insulin doses that aim to bring glucose back into the target range. This visualization exemplifies the complex dynamics and temporal dependencies that the model must capture to enable accurate and personalized glucose forecasting.

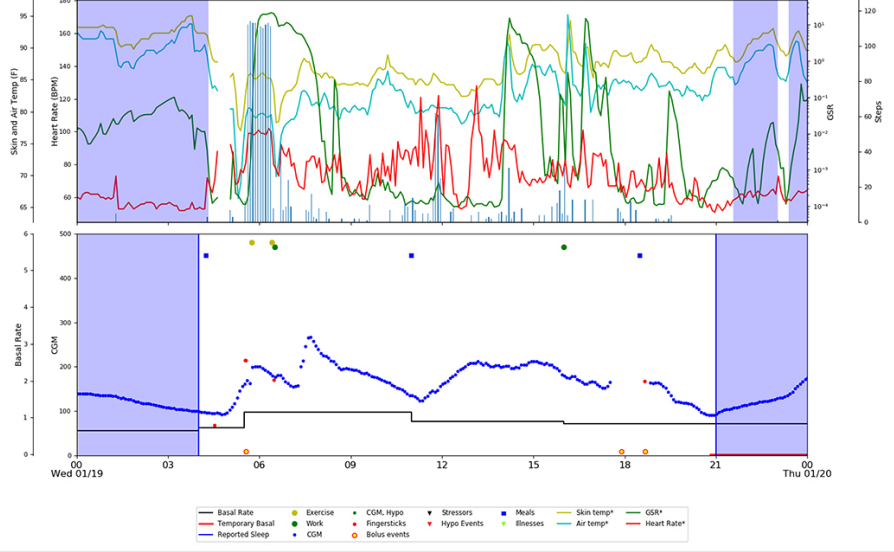


Figure 2: 24-hour measurements of one patient from the OhioT1DM Viewer [MB20].

3.2.3 MIMIC-IV:

The proposed model aims to generalize beyond diabetes forecasting to other medical applications, such as predicting disease progression in HFpEF patients. The MIMIC-III dataset [JPS⁺16] contains de-identified electronic health records (EHRs) of ICU patients, including vital signs (heart rate, blood pressure, oxygen saturation, etc.), medication records, and laboratory results. For this study, a subset of MIMIC-IV focusing on cardiovascular patients is utilized. The target variable, mortality risk, was divided into two groups: death within 30 days and death within one year. The exogenous variables include vital signs and laboratory values, while gender and comorbidities are used to split the data into multiple cohorts. By analysing the data in different cohorts, it is possible to get more specific and accurate counterfactual interventions.

3.2.3.1 Preprocessing

Following a preceding study[SBA⁺24], the International Classification of Diseases (ICD) codes were used for the initial preprocessing. The ICD-codes are a standardized international classification system used for the categorisation and encoding of diseases, symptoms, and associated health-related conditions. By using the appropriate ICD-9 and ICD-10 codes, as outlined in Table 1, the hospital admissions involving patients aged ≥ 18 with HFpEF as a primary diagnosis have been identified. Given that the diagnosis was based on

ICD codes, the clinical notes were analysed in order to validate the selection of patients. This was achieved through filtering the clinical notes on mentions of the left ventricular ejection fraction (LVEF) value, with a value of 50 and above being counted as a normal LVEF value. Some clinical notes only mentioned a normal or preserved LVEF value without a measured LVEF. These were also counted as normal LVEF values. Table 1 also shows the number of hospital admissions per diagnosis. The study sample consisted of 16122 individual hospitalizations with a suspected diagnosis of HFpEF. We had access to clinical notes for 11720 (72.7%) hospital admissions of which 4458 (38%) had an LVEF measurement reported. Of these, 3798 (85.2%) had an LVEF value $\geq 50\%$, and 400 (9%) had an LVEF $< 50\%$. An additional 260 (5.8%) admissions had mention of a normal or preserved LVEF. For these 4058 admissions, vital signs and laboratory values were available for 2432 (60%). It should be noted that in this instance, only the most recent admission of a patient who had previously been admitted with a similar diagnosis was taken into consideration. Prior admissions were incorporated into the analysis as a comorbidity, and after adding all laboratory values, vital signs and comorbidities, the resulting number of unique hospital admissions is 2113 with 1845 unique patients.

Diagnosis	ICD Code	Frequency
Unspecified diastolic (congestive) heart failure	I5030	38
Diastolic heart failure, unspecified	42830	69
Acute diastolic (congestive) heart failure	I5031	84
Acute diastolic heart failure	42381	180
Chronic diastolic (congestive) heart failure	I5032	256
Chronic diastolic heart failure	42382	453
Acute on chronic diastolic (congestive) heart failure	I5033	426
Acute on chronic diastolic heart failure	42833	623

Table 1: The corresponding Diagnosis and ICD-9 and ICD-10 codes for HFpEF.

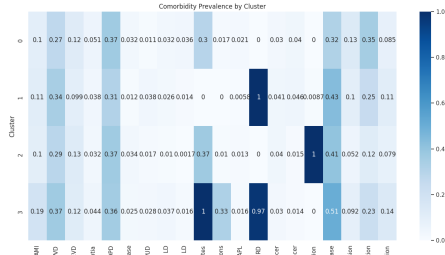
The extracted features are listed in table 2, split into four categories. These are the targets (Death within 30 days, Death within 1 year) as well as vital signs and laboratory values, comorbidities, and gender. Prior admission was included in the list of comorbidities, since it is here used as a comorbidity for the clustering and not the forecasting.

Feature types	Specifics	Occurrences
Targets	Death within 30 days	136
	Death within 1 year	150
Vital signs and laboratory values	BMI	1845
	Heart Rate	1845
	SpO2	1845
	Diastolic BP	1837
	Systolic BP	1837
	Temperature	1829
	Creatinine	1825
	Sodium	1825
	Bicarbonate	1823
	Hemoglobin	1814
	WBC Count	1814
	Platelet Count	1812
	Troponin	958
	NT-proBNP	76
Comorbidities	Diabetes	798
	RD	768
	Coronary Artery Disease	761
	COPD	658
	Hypertension	599
	PVD	577
	Atrial Fibrillation	424
	AMI	230
	CEVD	218
	Prior Admission	188
	Pulmonary Hypertension	170
	Diabetes + Complications	156
	Dementia	75
	Cancer	65
	Metastatic Cancer	50
	Rheumatoid Disease	50
	Mild LD	46
	PUD	40
	Moderate/Severe LD	30
	HP/PAPL	27
Sex	Male	773
	Female	1072

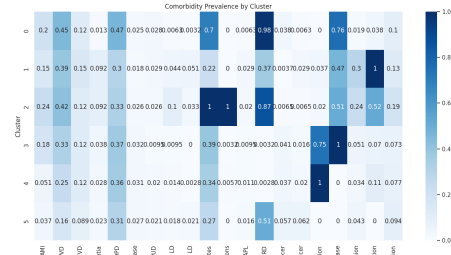
Table 2: All features of the MIMIC data divided by feature types: Target, Vital signs and laboratory values, Comorbidities, and Gender.

3.2.3.2 Clustering

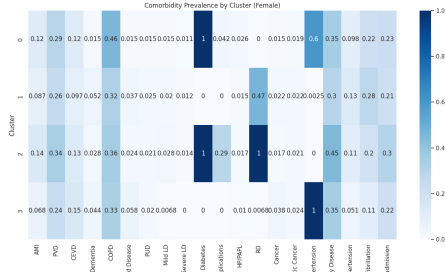
The MIMIC data does not only include a wide range of features, but also a great number of comorbidities. Since these comorbidities can influence the chances of HFpEF, it is important to include these in the analysis. In this study, the comorbidities were used to cluster the patients, to get more specific counterfactuals. For this, the data was split according to patients having similar comorbidities to get factual cohorts. Another split was made by dividing the data on gender, since HFpEF is more prevalent in female patients and might need different treatment. Figure 3 shows the prevalence of different comorbidities of the MIMIC patients in the clusters, divided into four sub figures depending on the clustering coefficient and the gender. Figure 4 shows the Principal Component Analysis (PCA) of the different comorbidities, again divided into the same subgroups.



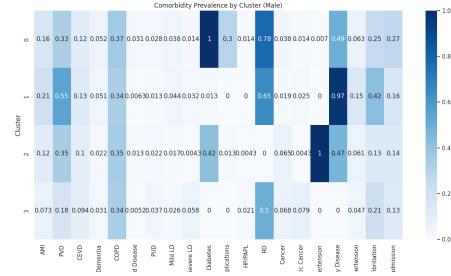
(a) Comorbidities in 4 clusters.



(b) Comorbidities in 6 clusters.



(c) Comorbidities for female patients in 4 clusters.

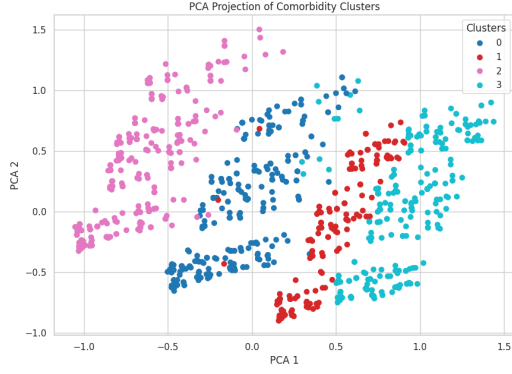


(d) Comorbidities for male patients in 4 clusters.

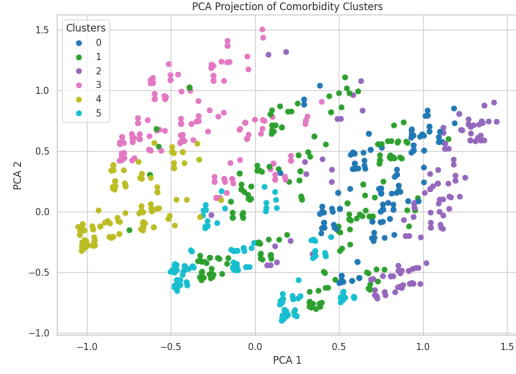
Figure 3: Prevalence of different comorbidities of the MIMIC patients in the clusters. There are 4 subgroups, clustering by comorbidity with $k = 4$, clustering by comorbidity with $k = 6$, clustering only the female patients with $k = 4$ and clustering only the male patients with $k = 4$.

3.3 Experiments

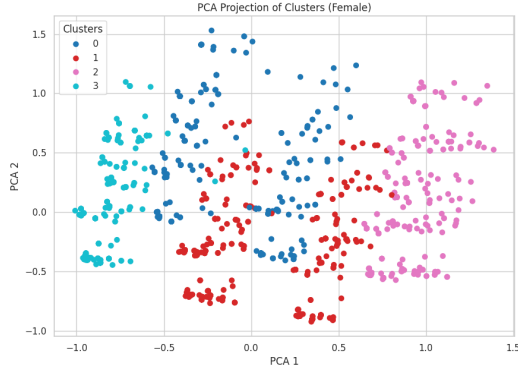
The model can be split into two main parts. Initially a multivariate forecasting model is used to make a first forecast for both the target variable and the exogenous variables. This forecast is then used for the second part, where different regression models are used to change the exogenous and target variable to get the desired outcome. For the forecasting part we used GRU and NBEATS and for the second part we used four different kinds of models, like a statistical (SARIMAX), a regression based (OLS), and two different deep learning based (GRU and NBEATSx) models.



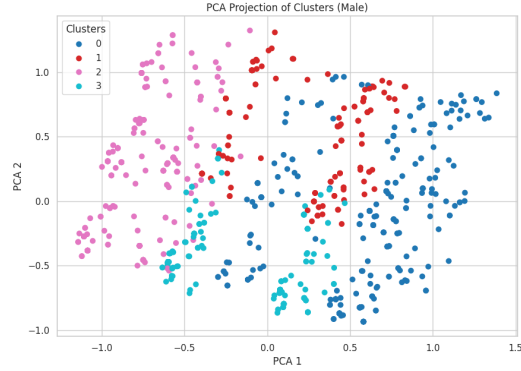
(a) PCA clustering in 4 clusters.



(b) PCA clustering in 6 clusters.



(c) PCA clustering for female patients in 4 clusters.



(d) PCA clustering for male patients in 4 clusters.

Figure 4: Principal component analysis (PCA) clustering of different comorbidities of the MIMIC patients. There are 4 subgroups, clustering by comorbidity with $k = 4$, clustering by comorbidity with $k = 6$, clustering only the female patients with $k = 4$ and clustering only the male patients with $k = 4$.

3.3.1 Multivariate Forecasting

For the multivariate forecasting task, two deep learning architectures were implemented and evaluated: (1) a 2-layer Gated Recurrent Unit (GRU) model, and (2) a 4-layer Neural Basis Expansion Analysis for Time Series (N-BEATS) model. The GRU model consisted of two stacked layers, each comprising 200 hidden units, followed by a linear output layer to produce the forecast. The N-BEATS model was configured with four fully connected layers, integrating both backcast and forecast blocks, and concluded with a linear output head to reconstruct future values. This design allows the model to capture both short-term patterns and longer-range temporal dependencies effectively.

To prevent overfitting, early stopping was employed in both architectures with a patience of 10 epochs, and training was conducted using a fixed learning rate of 0.001. The models were trained on varying backward input horizons and forecast windows, allowing a thorough investigation of how different historical contexts influenced future prediction accuracy.

Model performance was evaluated using two standard error metrics: symmetric Mean Ab-

solute Percentage Error (sMAPE) and Root Mean Squared Error (RMSE). These metrics were selected to balance sensitivity to outliers (RMSE) with scale-invariant accuracy (sMAPE), ensuring a robust evaluation of predictive performance. Lower values for both metrics indicated better forecasting quality. Model performance was assessed using four complementary metrics: Symmetric Mean Absolute Percentage Error (sMAPE), Root Mean Squared Error (RMSE), Accuracy, and F1 Score.

- **Symmetric Mean Absolute Percentage Error (sMAPE):** sMAPE measures the relative accuracy of forecasts by comparing the absolute difference between predicted and actual values to their average magnitude. It is scale-invariant, making it suitable for comparing performance across different ranges. Lower sMAPE values indicate higher predictive accuracy.
- **Root Mean Squared Error (RMSE):** RMSE quantifies the square root of the average squared differences between predicted and actual values. It is sensitive to large errors (outliers) and provides insight into the magnitude of typical prediction errors. Lower RMSE values reflect better model performance.
- **Accuracy:** Accuracy represents the proportion of correctly predicted instances (both positives and negatives) out of all predictions. It provides a straightforward measure of overall model correctness, particularly relevant in binary classification tasks.
- **F1 Score:** The F1 score is the harmonic mean of precision and recall. It balances the trade-off between false positives and false negatives, offering a robust evaluation for imbalanced classification scenarios. A higher F1 score indicates better balance and reliability in predicting the positive class.

These four metrics were chosen to provide a comprehensive evaluation framework. To assess the continuous forecasting performance as necessary for the OhioT1DM and SimGlucose datasets, sMAPE and RMSE are used. Accuracy and F1 Score specifically evaluate classification performance, particularly important for binary outcomes as found in the MIMIC dataset. In all cases, lower sMAPE and RMSE values, along with higher Accuracy and F1 scores, indicate better model quality.

After thorough evaluation across multiple configurations, the model with the lowest average error on all validation sets was selected as the baseline for subsequent counterfactual analysis. This selection ensured that counterfactual explanations were derived from the most reliable and accurate forecasting model available.

[**TODO:** Extend on method]

3.3.2 Counterfactual Generation

To evaluate our counterfactual generation approach, we conducted experiments on two clinical prediction tasks: glycaemic forecasting for Type 1 Diabetes and survival prediction for Heart Failure with Preserved Ejection Fraction (HFpEF). For each task, we applied domain-specific thresholds to define risk groups and designed personalized counterfactuals that suggest minimal but effective interventions.

The models used for the counterfactual generation span both traditional statistical methods and deep learning architectures, such as SARIMAX, OLS, GRU, and N-BEATSx.

Each model was adapted to support input optimization using either gradient-based back-propagation (for differentiable architectures) or finite-difference approximation (for black-box models).

3.3.2.1 SARIMAX

We used the Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX) model to capture seasonal patterns and time dependencies in glucose data, while including outside factors like insulin doses and carb intake. For each data sample, we fit a SARIMAX with autoregressive order $p = 1$, differencing degree $d = 0$, and moving average order $q = 0$ using maximum likelihood estimation via the statsmodels library. Hyperparameters were chosen based on AIC scores. When generating counterfactuals, we adjusted input exogenous variables iteratively to keep predicted glucose within set bounds. We calculated gradients approximately using finite differences on the forecast function of the fitted model. To make changes more realistic, we applied custom weighting focused on meal-related features during optimization.

3.3.2.2 OLS

We also applied Ordinary Least Squares (OLS) regression similarly to SARIMAX, modeling the linear relationship between lagged inputs (including interaction terms like insulin and carbs) and glucose levels. Each sample had its own OLS model. Counterfactual optimization followed the same process as SARIMAX, by using finite difference gradients to guide input changes, constrained by feature-specific limits and weighted towards meal-related inputs to keep the results physiologically plausible.

3.3.2.3 GRU

The Gated Recurrent Unit (GRU) model had two recurrent layers with 100 units each, followed by a dense layer outputting predictions. It was trained on multiple input signals like carbs, insulin, and physical activity, with dropout of 0.2 to avoid overfitting. Training used the Adam optimizer with a learning rate of 0.001, and early stopping based on validation loss. For counterfactuals, we used TensorFlow’s automatic differentiation to get exact gradients of outputs with respect to inputs. This allowed efficient gradient-based optimization of input features under clipping limits and with custom weighting focused on meal times. We stopped the optimization once predicted glucose stayed within desired ranges or after a max number of steps.

3.3.2.4 NBEATSx

NBEATSx is a version of the N-BEATS model designed to explicitly incorporate external inputs like carbs, insulin, and activity into its trend and seasonal components. It used fully connected residual blocks with 128 hidden units and was trained to minimize mean absolute error. Training used the Adam optimizer with a 0.001 learning rate. Like the GRU, counterfactual generation was done with gradient-based optimization using TensorFlow’s autodiff, constrained by clipping and guided by meal-focused weighting. This allowed the model to capture complex nonlinear patterns while making it easier to interpret how exogenous inputs affect glucose predictions during counterfactual analysis.

3.3.2.5 Parameters for the Diabetes Data (OhioT1DM, SimGlucose)

For the diabetes datasets, test samples were split into subgroups based on predicted glucose levels over a 30-minute forecast horizon:

- Hyperglycemia group: samples with any predicted value $t_{n+i} \geq 180$ mg/dL.
- Hypoglycemia group: samples with any predicted value $t_{n+i} \leq 70$ mg/dL.

For each group, we randomly selected 100 test samples (or fewer if not available) for counterfactual generation. The same process was repeated across datasets with dataset-specific bounds: for SimGlucose, we used [80, 160] mg/dL; for OhioT1DM, [70, 180] mg/dL.

3.3.2.6 Parameters for the HFpEF Survival Prediction

In the HFpEF dataset, binary predictions over a 24-step horizon were used to classify samples as high-risk (non-survival) if more than 50% of the time steps predicted a non-survival event (label 1). A total of 100 samples were selected from each group:

- Non-survival group: $\sum_{i=1}^{time} t_{n+i} \geq time/2$
- Survival group: $\sum_{i=1}^{time} t_{n+i} < time/2$

[TODO: check]

3.3.2.7 Counterfactual Generation Procedure

We followed a consistent three-step pipeline:

- Bound Generation: For continuous targets (OhioT1DM and SimGlucose), we constructed soft upper and lower bounds using a polynomial interpolation function transitioning from the current forecast to a desired safe target over a fixed number of time steps ($S = 24$ for OhioT1DM, $S = 20$ for SimGlucose). For HFpEF, since the target is binary, no such bounds were applied.
- Optimization: Exogenous inputs (e.g., carbohydrate intake, insulin dose, clinical variables) were perturbed to minimize a custom loss function. For differentiable models (GRU, N-BEATS), gradients were obtained via backpropagation. For non-differentiable models (ARIMA, Linear), finite-difference approximations were used. For HFpEF, optimization aimed to flip the predicted class while minimizing input changes, without any bound constraints.
- Loss: The loss balanced moving predictions toward the target class or within bounds for continuous targets and minimizing the magnitude of input perturbations, where feature-specific weights emphasized intervention-relevant variables (e.g., insulin, carbohydrates for glucose; clinical features for HFpEF) to encourage realistic and meaningful counterfactuals.

All inputs and targets were scaled using per-patient normalizers to preserve physiological interpretability. Historical context (e.g., meal history, past glucose trends) was retained to ensure the counterfactuals remained plausible within each individual’s recent clinical trajectory.

[TODO: Extend on method, add more about mimic]

3.4 Evaluation Metrics

[**TODO:** add hfpef] To evaluate the quality of the counterfactual interventions, multiple evaluation metrics were implemented. First, traditional forecasting performance was measured using Root Mean Squared Error (RMSE) and Symmetric Mean Absolute Percentage Error (sMAPE), giving a quantitative assessment of prediction accuracy across the forecast horizon. For the evaluation of the generated counterfactuals, several additional metrics were introduced. These have been applied to ensure that the newly generated data for the exogenous variables is realistic, plausible and applicable.

To evaluate the capability of the optimizer in manipulating exogenous variables to bring predicted blood glucose (BG) levels within a desired target range, we designed an experiment using a recurrent model trained on real-world diabetes management data. Specifically, we used a GRU-based forecasting model trained on the OhioT1DM dataset to predict future BG levels, with associated exogenous variables including basal insulin, bolus insulin, carbohydrate intake, and exercise intensity.

Once the base model was trained, we applied a constrained optimization procedure to adjust the exogenous variables over a prediction horizon such that the predicted BG values fall within a physiologically reasonable range. The optimization is performed with the goal to minimize the deviation of predicted BG from the original forecast while forcing it to lie within a specified desired range (e.g., 70–180 mg/dL), as well as to limit the extent of changes made to exogenous variables (e.g., insulin or carbs) to maintain physiological and behavioural plausibility.

3.4.1 Magnitude and Sparsity of Exogenous Variable Changes

The average value of change calculates the mean absolute difference between original and counterfactual exogenous variables, averaged over all samples, time steps, and features. It reflects the overall magnitude of adjustments required. By determining the proportion of exogenous input entries that underwent modification, we assess the sparsity of the intervention, providing insight into whether changes are targeted or widespread across the input profile.

3.4.2 Severity of change

The severity metric measures changes normalized by the original variability of each exogenous feature, enabling assessment of whether modifications are within a physiologically reasonable range. For this, we employ Local Outlier Factor (LOF) scores to identify whether the counterfactual exogenous profiles exhibit atypical or extreme deviations from the original data distribution, highlighting potential concerns about the realism of the suggested interventions.

3.4.3 Fitting of Predictions within Bounds

The extent to which counterfactual BG predictions fall within the desired range is quantified by the percentage of time points inside the bounds and the root mean squared error (RMSE) of any violations outside these limits. This metric directly evaluates the success of the optimization in achieving clinically safe target BG levels.

3.4.4 Comparison to Healthy Exogenous Profiles

To ensure biological and clinical plausibility, the optimized exogenous variables are compared against reference exogenous profiles derived from healthy patient data. Both raw and normalized Euclidean distances, as well as cosine similarity metrics, quantify the alignment of counterfactual inputs with typical healthy patterns, supporting qualitative evaluation of intervention realism.

3.4.5 Interpretation

This multi-faceted evaluation framework provides a detailed and nuanced understanding of how effectively the optimization generates actionable and safe counterfactual interventions. The ideal outcomes achieve a careful balance between several important, sometimes competing, goals. First, they bring predicted blood glucose levels within the desired target range without causing the forecasts to deviate excessively from the original values, ensuring the adjustments remain realistic and consistent with the patient’s typical glucose patterns. At the same time, the changes to exogenous variables, such as insulin dosages or carbohydrate intake, are kept as small and infrequent as possible, since large or frequent modifications might be impractical or even unsafe in real-world settings. Additionally, the interventions maintain physiological plausibility by limiting the intensity of changes and avoiding outlier values that could indicate unrealistic or extreme behaviors. Clinical safety is also prioritized by strictly enforcing blood glucose predictions to stay within safe bounds, reducing the risk of hypo- or hyperglycemia. Finally, the adjusted exogenous inputs are aligned closely with patterns observed in healthy patients, making the suggested interventions not only effective but also clinically meaningful and believable. By assessing performance across all these dimensions, this evaluation approach highlights both the strengths and limitations of the counterfactual generation process, guiding future improvements and enhancing its practical value in diabetes management.

4 Results

4.1 Multivariate Forecasting

[TODO: explain chosen horizon, back horzion] To evaluate the performance of the multivariate forecasting and to ensure a realistic basis for counterfactual generation, both the GRU and the N-BEATS models were tested across multiple datasets and forecasting configurations. Models were trained on historical segments of the data and evaluated on subsequent future segments, allowing for a robust comparison between predicted and actual values. The forecasting quality for the two diabetes datasets was assessed using two standard metrics: Symmetric Mean Absolute Percentage Error (sMAPE) and Root Mean Squared Error (RMSE). sMAPE was chosen due to its scale-invariant properties, allowing for fair comparison across variables with different magnitudes. RMSE was included to capture sensitivity to larger errors, penalizing substantial deviations between predictions and true values. Together, these metrics provide a balanced view of model performance, addressing both relative and absolute error considerations. For the MIMIC-IV dataset, which involves binary survival prediction (death vs. survival), classification

metrics were more appropriate given the nature of the target variable. Specifically, Accuracy and F1-Score were employed. Accuracy provides a general measure of how often the model’s predictions match the true outcomes. However, due to the severe class imbalance in the MIMIC-IV dataset, where death events are comparatively rare, F1-Score was also reported to better capture the trade-off between precision and recall for the minority class. This combination of metrics ensures both the overall predictive reliability and the ability to identify critical but infrequent events are adequately evaluated.

Table 3 summarizes the results across the two diabetes datasets, prediction horizons, and back horizons. For the OhioT1DM dataset, N-BEATS consistently outperformed the GRU model in both sMAPE and RMSE across all configurations. Notably, with a back horizon of 12 and a forecast horizon of 6, N-BEATS achieved a sMAPE of 6.03 compared to GRU’s 6.28, and a lower RMSE of 14.15 versus 14.85 for GRU. The same pattern was observed for back horizon 6 and 24, with N-BEATS slightly outperforming GRU in all cases. For the SimGlucose dataset, N-BEATS again generally showed improved forecasting performance. While GRU slightly outperformed N-BEATS in sMAPE for the 20-step back, 5-step horizon configuration (0.3792 vs. 0.3867), N-BEATS achieved a substantially lower RMSE (0.8271 vs. 1.6770), indicating more accurate absolute predictions. This performance gap widened in the 40-back, 10-horizon configuration, where N-BEATS clearly surpassed GRU on both metrics.

[TODO: add mimic] Overall, these results demonstrate that the N-BEATS model generally offers superior performance across datasets and configurations, especially in terms of RMSE, making it a strong candidate for generating accurate multivariate forecasts and serving as a foundation for counterfactual analysis.

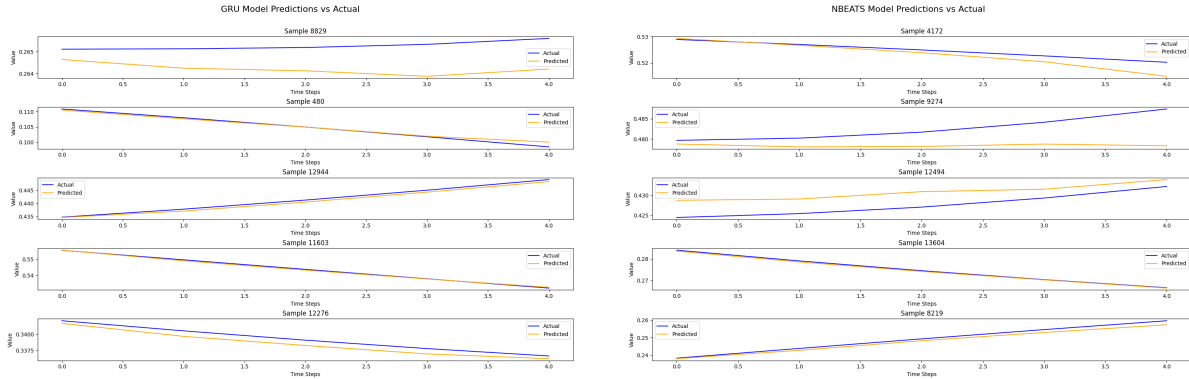
Table 3: Multivariate forecasting training metrics.

Dataset	Back Horizon	Horizon	Model	sMAPE	RMSE
OhioT1DM	12	6	GRU	6.2816	14.8471
			NBEATS	6.0328	14.1500
	24	6	GRU	6.1983	14.6856
			NBEATS	5.8003	13.7677
SimGlucose	20	5	GRU	0.3792	1.6770
			NBEATS	0.3867	0.8271
	40	10	GRU	1.5423	5.5572
			NBEATS	1.4303	3.0461

[TODO: rerun, recalculate results] To complement the quantitative evaluation of the multivariate forecasting models, several visualizations were created to illustrate how well the N-BEATS and GRU models track actual future values. These plots compare the predicted time series against the true values for randomly selected test samples from the datasets. Each subplot presents a single test sample, with time steps on the x-axis and the corresponding variable value on the y-axis. The actual values are shown in blue, while the predicted values are shown in orange. These plots offer a qualitative insight into the temporal alignment and amplitude accuracy of the forecasts, beyond what is captured by metrics like sMAPE or RMSE.

4.1.1 SimGlucose

[**TODO:** only 10,40, rest in appendix?] For a back horizon of 20 timesteps and a horizon of 5 timesteps, both the GRU and N-BEATS models operate on a relatively short forecast horizon, making it easier to capture the trend with high fidelity. As shown in Figure 5, N-BEATS predictions closely align with the actual values across multiple samples, reflecting low forecasting error. Minor deviations appear mostly at the end of the forecast window. While the trend direction for the GRU predictions is usually correct, the model exhibits slightly larger prediction drift compared to N-BEATS, especially in sharper transitions. Still, GRU maintains reasonable short-term accuracy. For a back horizon of 40 timesteps and a horizon of 10 timesteps, the forecast horizon is increased, making the prediction task more difficult. Figure 6 shows the N-BEATS model’s ability to handle long-range dependencies. Although small prediction lags and amplitude mismatches occur, the model captures the overall progression well, preserving directionality in most sequences. When looking at the GRU predictions under the same setup, the model struggles a bit more with extended forecasts, often showing divergence from actual trajectories, particularly toward the final time steps. These deviations are consistent with the slightly higher RMSE reported for this configuration.

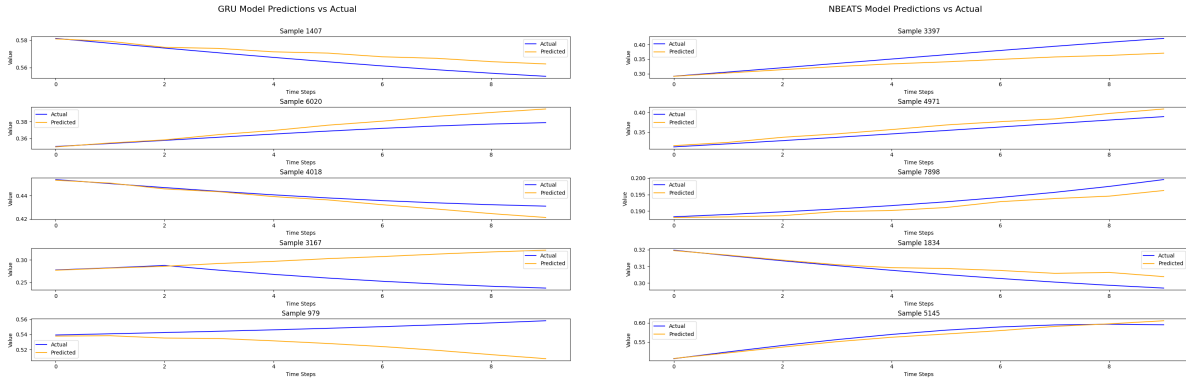


(a) Results of the multivariate forecasting using GRU. (b) Results of the multivariate forecasting using NBEATS.

Figure 5: Results of the multivariate forecasting for the SimGlucose dataset with back horizon = 20 and forecast horizon = 5, showing the accuracy of the forecasting.

4.1.2 OhioT1DM

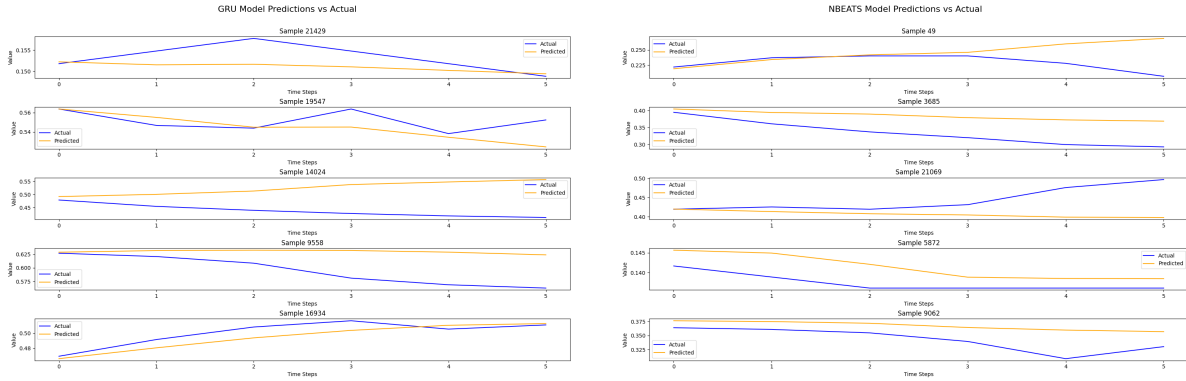
[**TODO:** only 6,24, rest in appendix?] Figure 7 compares N-BEATS and GRU predictions for a back horizon of 12 timesteps and 6 future timesteps. The N-BEATS model tracks the general shape and scale of the actual values effectively. Although slight under- or over-predictions occur, especially in rising or falling trends, the model remains closely aligned overall. In contrast, the GRU predictions tend to diverge slightly more in the later steps of the forecast horizon. GRU often preserves the direction of the trend but struggles with amplitude and slope consistency, leading to larger deviations as the prediction window progresses. With a back horizon of 24 timesteps and the same future timesteps, figure 8 depicts the performance of N-BEATS and GRU under a longer input sequence. The figure illustrates that N-BEATS maintains robust trend prediction, with only modest



(a) Results of the multivariate forecasting using GRU. (b) Results of the multivariate forecasting using NBEATS.

Figure 6: Results of the multivariate forecasting for the SimGlucose dataset with back horizon = 40 and forecast horizon = 10, showing the accuracy of the forecasting.

lag or dampening effects in some samples. The model adapts well to both upward and downward trajectories. The GRU model performs more variably under this longer back horizon. While the general trajectory is still often captured, the forecasts occasionally overreact or smooth out variations, resulting in lower precision at the end of the forecast horizon.



(a) Results of the multivariate forecasting using GRU. (b) Results of the multivariate forecasting using NBEATS.

Figure 7: Results of the multivariate forecasting for the OhioT1DM dataset with back horizon = 12 and forecast horizon = 6, showing the accuracy of the forecasting.

4.1.3 MIMIC-IV

[TODO: check, rewrite] For the MIMIC-IV dataset, the classification performance of GRU and N-BEATS models was evaluated using accuracy, precision, recall, and F1-score, with class-wise metrics reported for both 30-day and 1-year mortality prediction tasks, as seen in table 4. For 30-day mortality, the GRU model achieved an overall accuracy of 83.5%, while the N-BEATS model reached 91.6%. Despite these high overall accuracy scores, both models failed to correctly identify any instances of the "Died" class. Specifically, both GRU and N-BEATS models reported precision, recall, and F1-score values of

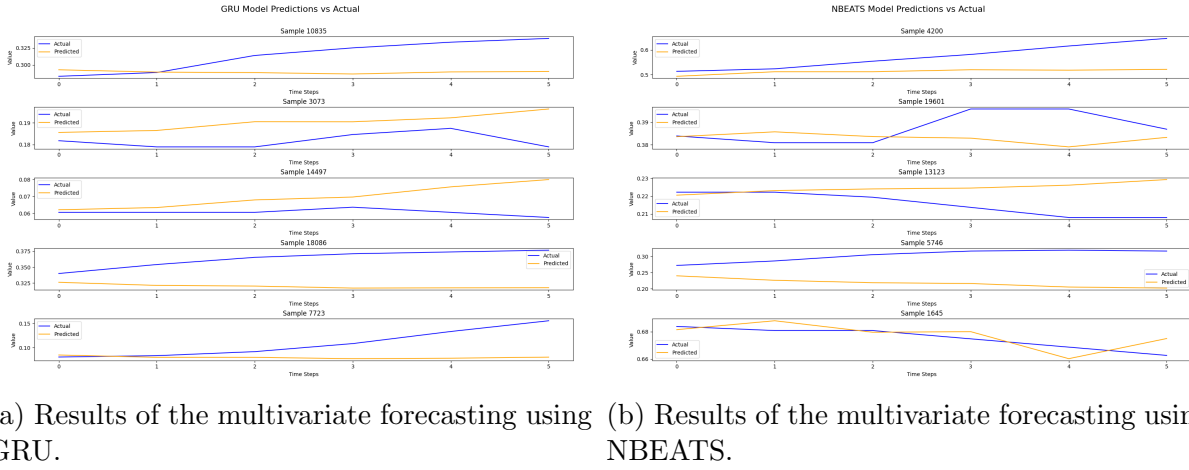


Figure 8: Results of the multivariate forecasting for the OhioT1DM dataset with back horizon = 24 and forecast horizon = 6, showing the accuracy of the forecasting.

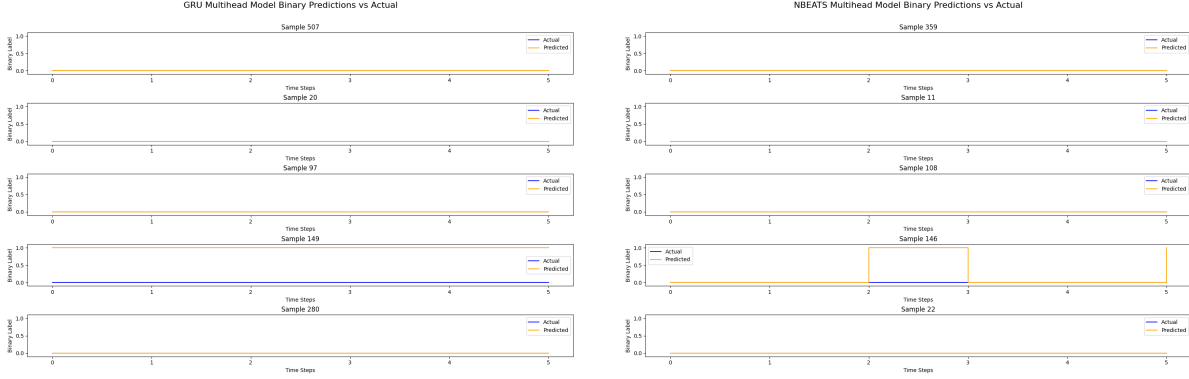
0.00 for the "Died" class, indicating that all predictions were biased toward the majority "Survived" class. This pattern persisted in the 1-year mortality prediction task, where both models again achieved high overall accuracy (90.6% for both), but with no correct predictions for the "Died" class. These results suggest that both models are biased towards the majority class, a common issue in imbalanced datasets. While overall accuracy may appear high, it does not reflect clinically important performance for detecting critical outcomes like patient death. Figure 33 compares the N-BEATS and GRU predictions for a

Table 4: Classification Metrics for GRU and N-BEATS Models Cluster 0

Model	Class	Precision	Recall	F1-Score	
GRU	30 days	Survived	0.91	0.86	0.88
		Died	0.00	0.00	0.00
		Overall Accuracy		0.7882	
	1 year	Survived	0.90	0.91	0.90
		Died	0.00	0.00	0.00
		Overall Accuracy		0.8235	
N-BEATS	30 days	Survived	0.92	0.97	0.94
		Died	0.00	0.00	0.00
		Overall Accuracy		0.8922	
	1 year	Survived	0.90	0.96	0.93
		Died	0.06	0.02	0.03
		Overall Accuracy		0.8706	

back horizon of 12 timesteps and 6 future timesteps and target variable 30-day mortality. Both models show a strong tendency to predict the majority class, which appears to be the negative (0) class in most samples. For example, in the plots for samples where the actual labels are all positive (1), such as Sample 53 in the N-BEATS model and Sample 71 in the GRU model, the predictions are consistently negative (0). This indicates a failure of both models to identify positive cases. Neither model demonstrates effective discrimina-

tion between classes at the individual sample level, as the predicted values remain flat and do not align with the actual binary changes over the time steps. The predictions appear to be largely constant across time steps, showing little to no responsiveness to actual positive events. This behavior suggests that both the N-BEATS and GRU models are biased toward predicting the majority class, likely due to class imbalance in the dataset. Despite potentially high overall accuracy, these models fail to capture the minority positive class, resulting in poor sensitivity and F1 scores for that class.



(a) Results of the multivariate forecasting using GRU. (b) Results of the multivariate forecasting using NBEATS.

Figure 9: Results of the multivariate forecasting for the MIMIC-IV dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

For complete results across all clusters and subgroups, refer to Appendix A.1.

4.2 Counterfactuals

4.2.1 OhioT1DM

Figure ?? presents an example of generated counterfactuals for the OhioT1DM dataset. It shows a short sequence of 6 timesteps where the predicted blood glucose (BG) level is approximately 155 mg/dL, which is lower than the original range of 180–190 mg/dL. This new BG level falls within the predefined desired bounds, indicating a plausible improvement. Notably, the generated exogenous variables, such as insulin doses, carbohydrate intake, and exercise intensity, differ considerably from the original values. These differences suggest actionable changes that may help achieve improved glycemic control.

To evaluate the realism of the generated exogenous variables, we searched for pairs of samples in the dataset where both the original and counterfactual blood glucose levels were similar. By comparing their exogenous features, we can assess whether the generated values are plausible within the real data distribution, ensuring that counterfactual recommendations are actionable and clinically meaningful.

For this analysis, we identified two samples of 6 timesteps each, with a minimal Euclidean distance of 2.0295 between their blood glucose target sequences. Their exogenous variables are compared in Figure 15. The observed differences highlight the variability in exogenous features that can still lead to similar blood glucose outcomes.

Additionally, Table 5 summarizes the Euclidean distances between exogenous features for the five closest pairs of samples, using both actual and normalized values. This quantitative measure helps to contextualize the variability seen in the visual comparison and supports the feasibility of the generated counterfactuals within real-world data ranges.

Feature	Pair 71, 48		Pair 27, 82		Pair 27, 11		Pair 68, 31		Pair 74, 75	
	Act	Norm	Act	Norm	Act	Norm	Act	Norm	Act	Norm
Basal Insulin	2.1575	0.5678	3.1525	0.6280	2.0002	0.2891	2.7798	0.3388	2.9895	0.5143
Bolus Insulin	19.4057	0.5012	21.2238	0.8798	19.8393	0.4116	10.8953	0.2978	15.0757	0.3838
Carbohydrates	100.5660	0.4907	80.1914	0.5888	125.0327	0.4353	31.7565	0.4943	143.2101	0.5476
Exercise Intensity	1.1705	0.5956	1.5558	0.9469	1.1187	0.5106	0.9448	0.2775	0.8671	0.7661

Table 5: Euclidean distances between exogenous features for the five closest target pairs in the OhioT1DM dataset, showing both actual and normalized values.

4.2.2 SimGlucose

Figure 16 present the results of counterfactual forecasting for the SimGlucose dataset using a GRU-based multivariate forecasting model. The aim was to identify alternative exogenous variable trajectories, specifically carbohydrate intake and insulin administration, that lead to improved blood glucose (BG) levels within a clinically desirable range. In the upper figure the predicted exogenous variables are compared against the original sequences. Both carbohydrate intake and insulin administration in the counterfactual scenario are noticeably reduced or even set to zero during most time steps. Carbohydrate intake increases only slightly at later time steps, while insulin remains consistently absent. This suggests that the model identifies minimal or delayed interventions as sufficient to guide BG levels toward healthier ranges, reflecting a preference for conservative adjustments.

The lower figure shows the impact of these counterfactual exogenous variables on BG trajectories. While the original BG levels remain relatively stable and slightly above the desired range, the predicted BG values stay well within the predefined target bounds (highlighted in green). The predicted trajectory is also smoother and lower, indicating improved glycemic control according to the model’s optimization.

Overall, these results demonstrate that the model effectively proposes simplified and realistic counterfactual adjustments in the SimGlucose environment. It prioritizes conservative intervention strategies, often minimizing both insulin administration and carbohydrate intake, while achieving BG outcomes that align more closely with clinical targets. This highlights the model’s potential applicability for decision support in artificial pancreas systems or glucose management recommendations.

4.2.3 MIMIC-IV

To evaluate how well our model can generate counterfactuals for clinical time series data, we tested several modelling approaches on the MIMIC-IV dataset, focusing specifically on patients with heart failure with preserved ejection fraction (HFpEF). The goal was to create realistic counterfactual trajectories that shift a patient’s predicted outcome from death to survival (binary target = 0), by making minimal and interpretable changes to exogenous clinical variables.

Figures 17 through 20 show examples of counterfactuals generated using four different models: SARIMAX, Ordinary Least Squares (OLS), GRU, and N-BEATS. In each case, the original values are plotted in blue, and the counterfactual values, those modified to achieve a survival outcome, are shown in yellow, across six time steps.

While all models share the same goal, altering the inputs just enough to flip the predicted outcome to survival while keeping the changes realistic, they approach the task in different ways, each with its own strengths.

SARIMAX (Figure 17) leans on temporal trends and smooth transitions. The adjustments it makes tend to be gradual and follow the original signal’s shape closely. It mainly modifies variables like Heart Rate, Bicarbonate, and Troponin T, all of which are clinically important in HFpEF. At the same time, it leaves others, such as BMI and Creatinine, mostly untouched, suggesting a conservative approach that focuses only on what’s most relevant.

OLS (Figure 18) takes a more direct, linear route to modifying inputs. While this makes the changes easy to understand, they can sometimes appear abrupt or less physiologically natural, especially for variables that typically show non-linear patterns (e.g., SpO2 or Platelet Count). OLS still gets the job done in terms of outcome flipping, but its results can be less smooth or realistic compared to the time series-specific models.

GRU (Figure 19) offers more flexibility thanks to its ability to capture complex temporal relationships. The counterfactuals it produces tend to preserve the overall shape of the original sequence while introducing focused changes to the most influential variables, like Hemoglobin, Temperature, and Heart Rate. GRU’s changes are more dynamic than SARIMAX’s but still maintain temporal coherence, making it well-suited for the variability found in clinical settings.

N-BEATS (Figure 20) also performs well, especially when it comes to maintaining both short- and long-term trends in the data. It introduces meaningful, smooth modifications to key features such as NT-proBNP and Systolic Blood Pressure, staying true to both clinical relevance and the natural flow of the time series. Its architecture allows for a high level of nuance in how changes are applied across time.

One notable pattern across all models is that certain features, such as BMI, Creatinine, and Platelet Count, tend to remain unchanged. This suggests that the models aren’t making arbitrary edits but are instead focusing on variables that actually influence the outcome. That kind of selective intervention supports interpretability and builds trust in the results.

In short, while all four models are capable of generating plausible counterfactuals that achieve the intended goal, they vary in how they balance smoothness, realism, and intervention strength. SARIMAX sticks to subtle, trend-based changes; OLS is simple and straightforward but sometimes rough; GRU introduces more flexible, targeted edits; and N-BEATS strikes a strong balance between structure and adaptability. These differences highlight the versatility of counterfactual reasoning for clinical time series and point to exciting opportunities for personalized intervention support in healthcare.

4.3 Evaluation Metrics

4.3.1 Model Comparison

Table 6 presents the average prediction error metrics across all samples and time steps for each evaluated model. The GRU and N-BEATS models significantly outperform SARIMAX and OLS in all prediction accuracy metrics. N-BEATS achieves the lowest MAPE (40.65%), indicating better proportional accuracy, while GRU attains the lowest MAE and RMSE.

Table 6: Prediction Deviation Metrics Across Models

Model	MAE	RMSE	Max Dev	MAPE (%)
GRU	50.67	50.94	203.47	53.35
N-BEATS	52.41	56.39	234.97	40.65
SARIMAX	111.25	111.48	812.03	116.53
OLS	116.61	116.85	797.12	123.57

Table 7 summarizes each model’s ability to generate predictions within the desired blood glucose bounds. GRU again performs best with 43.5% of predictions in range and minimal violation area. Despite N-BEATS’ low MAPE, its bound compliance is particularly poor (0.8%), with a large violation area—indicating that low absolute error does not necessarily translate into physiological safety or effectiveness.

Table 7: Bound Compliance and Violation Metrics

Model	In Bound (%)	Mean Violation	Violation Area
GRU	43.5	5.59	2944.51
N-BEATS	0.8	80.60	48170.33
SARIMAX	2.7	87.42	52421.28
OLS	5.7	86.26	51689.86

Table 8 compares the magnitude of changes to exogenous inputs required by each model. GRU introduces the smallest average interventions across all features, with changes remaining under 7%. In contrast, SARIMAX and OLS generate drastic changes—exceeding 200% on average—suggesting poor optimization stability or unrealistic control strategies. N-BEATS strikes a balance, introducing moderate exogenous adjustments (around 15%).

Table 8: Mean Percentage Change in Exogenous Inputs

Model	Basal Insulin	Bolus Insulin	Carbs	Exercise
GRU	6.3%	6.2%	5.8%	6.3%
N-BEATS	16.7%	14.3%	15.0%	15.4%
SARIMAX	211.7%	210.9%	211.8%	210.7%
OLS	200.5%	199.6%	200.5%	199.4%

4.3.2 Feature-wise Euclidean Distances

Table 9 summarizes the average Euclidean distances between original and counterfactual feature values, presented both in raw and normalized forms to account for feature scale differences.

Table 9: Average Euclidean Distances per Feature

Feature	Avg Euclidean Distance	Normalized Euclidean Distance
Basal Insulin	1.2356	4.2353
Bolus Insulin	7.1287	4.4934
Carbohydrates	59.2930	4.2498
Exercise Intensity	0.5816	4.1659

As observed, carbohydrates exhibit the largest raw magnitude of change, while normalized distances indicate comparable scale-adjusted changes across all features.

4.3.3 Average Change Per Feature

Table 10 shows the average magnitude of modification applied to each feature in the counterfactual generation process.

Table 10: Average Change per Feature

Feature	Avg Change
Basal Insulin	0.4400
Bolus Insulin	2.5774
Carbohydrates	21.0900
Exercise Intensity	0.2076

The carbohydrates feature undergoes the largest average adjustment, suggesting it plays a key role in driving counterfactual changes.

4.3.4 Healthy Patient Comparison

[TODO: add graph] The following table 11 presents detailed Euclidean distances between selected original patient samples and their closest healthy patient counterfactuals, for both raw and normalized values, per feature.

Figure 21 shows the corresponding plots for pair 5, to visualise how the counterfactuals differ from a comparative patient sample.

Across all pairs, bolus insulin and carbohydrates exhibit the largest raw Euclidean distances, which is consistent with the trends observed in the previous results. Although exercise intensity shows smaller changes in raw magnitude, its normalized distances reveal that these changes are still significant relative to the feature’s scale. The target Euclidean distances between the pairs range from 1.72 to 2.42, indicating a relatively close alignment between the original patient samples and their predicted healthy counterparts. Additionally, there is noticeable variability in the normalized distances across features, reflecting different adjustment requirements for each patient-feature combination.

Table 11: Euclidean Distances for Healthy Patient Comparisons

Pair	Target	Basal Insulin		Bolus Insulin		Carbohydrates		Exercise Intensity	
	Euclidean	Act.	Norm.	Act.	Norm.	Act.	Norm.	Act.	Norm.
1	1.7209	0.8084	2.7711	7.6929	4.8490	48.4378	3.4718	0.3876	2.7764
2	2.1164	0.7179	2.4608	7.7761	4.9014	49.4722	3.5459	0.3764	2.6960
3	2.2299	1.5801	5.4161	11.6026	7.3134	75.4905	5.4108	0.9714	6.9587
4	2.3459	1.9827	6.7964	7.9109	4.9864	89.2984	6.4004	0.4430	3.1732
5	2.4201	1.4544	4.9852	5.0520	3.1844	63.4633	4.5487	0.3887	2.7841

4.3.5 Overall Counterfactual Evaluation Metrics

Table 12 compiles the key global metrics assessing the counterfactual set’s properties, feasibility, and similarity to healthy samples.

Table 12: Global Counterfactual Evaluation Metrics

Metric	Value
Fraction of Values Changed	0.9979
Average Z-Score Change	37.9923
Average Local Outlier Factor (LOF)	-1.0000
Percent Within Bounds	41.6667%
Violation Root Mean Square Error	3.5656
Mean Absolute Difference to Healthy	6.8727
Cosine Similarity to Healthy	0.9085

4.3.5.1 Interpretation

Nearly all feature values are changed (99.8%), indicating extensive counterfactual modifications. The large average z-score change, approximately 38, signifies substantial deviations relative to the underlying feature distributions. The LOF score of -1.0 suggests that the counterfactuals lie within the natural data manifold, meaning they are considered inliers. However, only 41.7% of the counterfactuals satisfy the imposed bounds, which points to potential feasibility concerns. The violation RMSE quantifies the extent of these constraint violations and highlights areas where improvements are needed. Despite this, the high cosine similarity of 0.91 indicates that the counterfactuals maintain a close directional alignment with the healthy samples. Finally, the mean absolute difference of 6.87 shows a moderate absolute distance from the healthy reference points.

The evaluation reveals that carbohydrates and bolus insulin features undergo the most significant changes in producing counterfactuals. While the counterfactuals largely reside within the data manifold, only a portion satisfy all constraints, suggesting room for improvement in enforcing feasibility. Overall, the counterfactuals demonstrate promising similarity to healthy samples but warrant careful clinical validation due to the scale of feature modifications.

4.3.6 Blood Glucose Prediction Performance

The blood glucose prediction deviations were evaluated over 100 samples and 6 time steps. Table 13 summarizes the MAE, RMSE, and MAPE metrics per time step. The Mean Absolute Error (MAE) remained relatively stable across the time horizon, ranging from 49.94 to 51.49, with the highest error observed at Time 4. Root Mean Squared Error (RMSE) values followed a similar trend, varying between 64.88 and 66.91. The Mean Absolute Percentage Error (MAPE) was consistently high, between 52.22% and 54.37%, indicating substantial relative errors despite moderate absolute deviations. The maximum observed deviation across all predictions was 203.47, suggesting occasional large outliers.

Table 13: Blood Glucose Prediction Deviation Metrics (per time step)

Time Step	MAE	RMSE	MAPE (%)
0	50.29	65.56	52.81
1	50.70	65.99	53.46
2	50.99	66.40	53.66
3	50.61	66.12	53.59
4	51.49	66.91	54.37
5	49.94	64.88	52.22
Overall Max Dev		203.47	

4.3.7 Compliance with Desired Blood Glucose Bounds

Table 14 details the percentage of predictions within desired blood glucose bounds, mean violation magnitude, and total violation area per time step. The percentage of predictions within the clinically desired bounds varied considerably over time. Early and late time points (timesteps 0, 3, 4, and 5) exhibited lower compliance, with in-bound percentages around 34–36%, whereas the middle time steps (timesteps 1 and 2) showed improved compliance, reaching up to 69% at timestep 2. Mean violation magnitudes were highest at timesteps 1 and 2, reflecting that when predictions fell outside bounds during these steps, the deviations were more severe. Total violation areas, representing cumulative out-of-bound magnitudes, were largest at timestep 0 and gradually decreased through timestep 4, before a slight increase at timestep 5. These results indicate temporal variability in the model’s ability to adhere to target glucose constraints, with better performance mid-horizon but challenges at the boundaries.

4.3.8 Exogenous Variable Adjustments Over Time

The exogenous variables, basal insulin, bolus insulin, carbohydrates, and exercise intensity, show differing magnitudes of adjustment throughout the time series, are summarized in Tables 15 to 18. For the basal insulin, the adjustments were minimal and stable across time, with MAE values between 0.06 and 0.08 and mean percentage changes consistently around 5–7%. Maximum deviations remained low (approximately 0.20 to 0.25 units), suggesting small but steady basal insulin modulations. In comparison, the bolus insulin values exhibited larger changes than basal insulin, with MAE increasing from 0.30 to 0.42 over

Table 14: Blood Glucose vs Desired Bounds (per time step)

Time Step	In Bound (%)	Mean Violation	Total Violation Area
0	35.0	11.93	775.71
1	52.0	12.10	580.70
2	69.0	13.50	418.36
3	34.0	5.90	389.17
4	36.0	5.43	347.79
5	35.0	6.66	432.79

time and percentage changes around 5–7%. Max deviations up to 1.09 units indicate moderate but significant bolus insulin adjustments. The largest exogenous changes occurred in carbohydrate intake, with MAE values rising from 2.33 to 3.75 and maximum deviations exceeding 11 units. The mean percentage change also increased over time, peaking at 7.15%, reflecting substantial dietary interventions, particularly in later time steps. For exercise intensity, the changes were minimal and consistent, with MAE around 0.03–0.04 and max deviations below 0.12. Mean percentage changes hovered near 6%, indicating stable but small modifications in exercise intensity. The prediction errors as well as the percentages of the targets that are in the bounds over time, can be seen in figure ??.

Table 15: Basal Insulin Changes (per time step)

Time Step	MAE	Max Dev	Mean % Change
0	0.06	0.25	5.04
1	0.07	0.24	6.25
2	0.07	0.20	6.69
3	0.07	0.22	6.43
4	0.08	0.24	6.73
5	0.07	0.21	6.47

Table 16: Bolus Insulin Changes (per time step)

Time Step	MAE	Max Dev	Mean % Change
0	0.30	0.86	5.24
1	0.34	1.09	5.79
2	0.37	1.07	6.00
3	0.37	1.06	6.32
4	0.42	0.92	6.81
5	0.42	1.05	6.97

Table 17: Carbohydrates Changes (per time step)

Time Step	MAE	Max Dev	Mean % Change
0	2.33	11.11	4.22
1	2.96	10.10	5.55
2	2.98	10.74	5.57
3	3.37	11.34	6.44
4	3.75	9.89	7.15
5	3.20	8.98	6.08

Table 18: Exercise Intensity Changes (per time step)

Time Step	MAE	Max Dev	Mean % Change
0	0.03	0.10	6.09
1	0.03	0.10	6.53
2	0.03	0.12	6.25
3	0.03	0.09	5.85
4	0.04	0.09	6.91
5	0.03	0.09	6.39

5 Discussion

5.1 Multivariate Forecasting

5.1.1 Results

[**TODO: überarbeiten, kontrollieren**] The multivariate forecasting experiments highlight key insights into the performance dynamics of sequence models across different datasets and configurations. Overall, the N-BEATS model demonstrated stronger forecasting capabilities compared to the GRU, as reflected in both the quantitative metrics (sMAPE and RMSE) and qualitative visualizations.

Across datasets such as OhioT1DM and SimGlucose, the N-BEATS model consistently outperformed GRU in RMSE, indicating improved absolute accuracy in forecasting. Even in configurations where GRU achieved slightly lower sMAPE scores (e.g., SimGlucose 20-back, 5-horizon), the N-BEATS model’s RMSE was substantially better, pointing to more reliable magnitude predictions even when percentage error might appear marginally higher. This is particularly relevant in domains like blood glucose forecasting, where the scale of the error directly impacts clinical decisions. The GRU model, on the other hand, often captured general trends and directions effectively but struggled with capturing exact amplitudes, particularly in extended horizon forecasts. Its recurrent structure may limit its ability to retain long-range dependencies, especially when faced with sparse or irregular temporal dynamics.

The impact of forecast horizon length is clearly observable, as the prediction window increases (e.g., from 5 to 10 steps), both models encounter greater difficulty maintaining accuracy, but N-BEATS shows better resilience to degradation. The model’s architectural design, which decomposes the forecasting problem into backcast and forecast components,

seems better suited for extrapolating patterns over longer horizons. The choice of back horizon, so how far back in time the model looks, also influences performance. Increasing the back horizon generally benefits N-BEATS more than GRU, likely due to its ability to effectively process longer sequences through its feedforward blocks. For example, in the OhioT1DM dataset, moving from a back horizon of 12 to 24 improved both sMAPE and RMSE in N-BEATS, whereas GRU showed more modest gains.

Dataset complexity plays a role in forecasting difficulty. The SimGlucose dataset, which is synthetic and well-structured, yields much lower errors compared to the real-world OhioT1DM data. This highlights the challenge of modelling patient-specific behaviors and noisy real-world measurements. Nevertheless, the relative performance gains of N-BEATS persist across both simulated and real datasets, reinforcing the robustness of its design.

Visualizations also confirmed the metric-based assessments, as N-BEATS forecasts track ground truth values more closely across time, particularly in maintaining amplitude and responding to abrupt trend changes. GRU’s forecasts tend to exhibit smoother transitions and sometimes dampened responses to sharp changes. These patterns match the sMAPE and RMSE trends, suggesting the chosen metrics are representative of the qualitative forecasting quality. Given that accurate forecasting is foundational to generating meaningful counterfactuals, these results justify the selection of the N-BEATS model as the primary forecasting component in the counterfactual pipeline. Its superior RMSE performance ensures that forward projections are grounded in realistic, data-consistent trajectories, which is crucial when optimizing exogenous variables to achieve desired outcomes. Additionally, its architectural stability over longer input sequences allows for greater flexibility in back horizon design.

While these findings strongly support N-BEATS for multivariate forecasting, experiments on the MIMIC-IV classification task introduce an important caveat. Despite achieving high overall accuracy in predicting 30-day and 1-year mortality, both N-BEATS and GRU failed to identify any positive cases of the minority “Died” class—resulting in zero precision, recall, and F1-score for that outcome. This stark performance drop highlights the challenge of applying these sequence models, designed primarily for regression, to severely imbalanced classification problems.

This observation emphasizes that model suitability is highly context-dependent: while N-BEATS is well-aligned with continuous time-series forecasting, specialized techniques (e.g., class rebalancing, focal loss) may be necessary to adapt such architectures for classification tasks, particularly in healthcare scenarios where identifying rare events is critical. The discrepancy also reinforces the need to carefully select metrics and training strategies that match task-specific goals.

In conclusion, the experimental evidence supports the adoption of N-BEATS as the preferred forecasting model within the counterfactual generation framework. Its robustness across datasets and input configurations ensures that recommendations are based on realistic, accurate projections. At the same time, the MIMIC-IV results serve as a reminder that forecasting strengths do not necessarily translate to classification accuracy in all contexts, underscoring the need for task-aware model design.

5.1.2 Limitations

While the GRU and N-BEATS models demonstrate strong performance on aggregate evaluation metrics, a critical limitation was observed in their inability to detect the minority "Died" class. Both models predominantly focus on the majority "Survived" class, which skews performance metrics such as accuracy and mean squared error. This imbalance is especially problematic in clinical prediction settings where identifying high-risk patients is of paramount importance. The observed class imbalance inherently biases the learning process, causing the models to overlook rare but clinically significant outcomes. This limitation suggests that relying solely on conventional evaluation metrics may provide an incomplete picture of model utility in imbalanced clinical datasets. The results highlight the pressing need for incorporating alternative strategies, such as resampling techniques, class weighting, or cost-sensitive learning, to ensure more equitable predictive performance across all outcome classes. Without such interventions, these models may fail to provide actionable insights in real-world healthcare applications, where missing minority class instances can have serious implications.

5.1.3 Future Work

Addressing the limitations outlined above should be a central focus of future research. Several promising strategies could be explored to mitigate class imbalance effects and improve minority class detection. One approach involves implementing class weighting during model training, adjusting the loss function so that misclassifying instances of the minority "Died" class incurs a higher penalty. This adjustment would compel the models to pay greater attention to these critical cases.

Additionally, resampling strategies such as Synthetic Minority Over-sampling Technique (SMOTE), random under-sampling of the majority class, or hybrid resampling methods can be applied to rebalance the training dataset. Specialized loss functions, including focal loss or those optimized for metrics like Matthews correlation coefficient, may also enhance minority class prediction by explicitly addressing class distribution disparities.

Further investigation into model architecture adjustments could offer additional improvements. For instance, integrating attention mechanisms or developing hybrid models that combine sequence learning with anomaly detection frameworks may provide more nuanced representations of minority class instances. Threshold tuning and calibration techniques could also be leveraged to fine-tune decision boundaries and improve sensitivity to minority outcomes.

In conclusion, while the GRU and N-BEATS models show promise on general predictive tasks, realizing their full clinical utility requires targeted strategies to enhance minority class performance. Addressing class imbalance through these techniques will be essential for developing predictive models that are not only statistically robust but also practically valuable in healthcare decision-making contexts.

5.2 Counterfactuals

The results show that generating multivariate counterfactuals for time series forecasting is not only possible but also practically useful. By adjusting exogenous variables while preserving the natural shape and rhythm of the data, the counterfactual sequences successfully reach desired target outcomes, such as keeping key variables within certain clinical

ranges, without introducing unrealistic values.

Both visual and quantitative evaluations suggest that the counterfactuals stay within the bounds of what we’d expect from real patient data. When compared with naturally occurring trajectories that lead to similar outcomes, the counterfactuals often follow different paths using different combinations of features. This reflects a core characteristic of multivariate time series: multiple input patterns can lead to similar predictions due to flexibility or redundancy in the system.

We also observed that the ”distance” between counterfactuals and nearby real samples helps highlight how different features can work together to drive a particular outcome. Instead of relying on a single-variable tweak, the models tend to adjust several features at once, much like how real clinical decisions often involve multiple interventions happening in tandem.

Taken together, these findings support the idea that multivariate counterfactual generation can be a valuable tool for time series forecasting tasks where interpretability and actionable insight are important. Moving forward, this work could be extended by applying domain-specific constraints, incorporating user feedback, or bringing in causal modelling to strengthen the relevance of generated interventions.

5.2.1 OhioT1DM

In the OhioT1DM results, the counterfactual sequence successfully brings BG levels down into a healthier range (around 155 mg/dL), starting from higher original values (180–190 mg/dL). This shift is achieved by changing key exogenous factors, particularly insulin dosage, carbohydrate intake, and exercise intensity. While these counterfactual adjustments differ noticeably from the original data, they remain within clinically plausible ranges, suggesting that the model identifies meaningful levers for improving glycemic control.

To evaluate how realistic these adjustments are, we compared the counterfactual inputs to real pairs of samples in the dataset with similar BG levels. These comparisons (visualized in Figure 15 and summarized in Table 5) reveal that quite different input patterns can still lead to similar BG outcomes. This variability supports the idea that there’s no single ”correct” intervention, multiple combinations of inputs can achieve the same result, and the counterfactuals generated here appear to fall within those bounds.

Overall, these results highlight the model’s ability to generate not just effective but also flexible and interpretable recommendations. The variability observed between similar BG outcomes further reinforces the value of multivariate approaches that consider a range of plausible paths rather than relying on single-variable tweaks.

5.2.2 SimGlucose

In the SimGlucose setting, the counterfactual sequences take a slightly different approach. The model tends to propose conservative strategies, minimizing both carbohydrate intake and insulin doses to improve BG levels. In many cases, insulin is reduced to zero, and carb intake is either flat or only slightly increased in later time steps. These subtle adjustments still lead to BG predictions that fall comfortably within the desired range, showing smoother and more stable glycemic trajectories.

This behavior reflects a model that’s tuned not just for accuracy but also for simplicity in intervention, something that could be especially valuable in real-world decision support

tools. For example, in artificial pancreas systems or diabetes management apps, recommending minimal yet effective interventions is often preferable to aggressive changes. Together, these findings demonstrate the potential for counterfactual generation in personalized glucose management. The models are able to suggest realistic adjustments tailored to the patient context while respecting physiological and behavioral boundaries. Whether through more varied strategies in OhioT1DM or conservative control in SimGlucose, both models showcase how counterfactual reasoning can enhance interpretability and support meaningful decision-making in clinical time series forecasting.

5.2.3 MIMIC

This study explored the use of counterfactual generation in clinical time series data, focusing on patients with heart failure with preserved ejection fraction (HFpEF) in the MIMIC-III dataset. The goal was to generate alternate trajectories for exogenous clinical variables that could change a predicted outcome from death to survival, in a way that remains realistic and interpretable.

All four modelling approaches (SARIMAX, OLS, GRU, and N-BEATS) were able to generate counterfactual sequences that followed physiological patterns while successfully flipping the model’s prediction. That said, each model approached the task in different ways, showing their unique strengths and limitations.

SARIMAX, a more traditional statistical model, produced gradual, trend-consistent changes that respected the structure of the original signal. This made the counterfactuals smooth and easy to interpret, though the model’s reliance on linear relationships limited its flexibility in capturing more complex or nonlinear dynamics.

OLS offered simplicity and interpretability but struggled with temporal consistency. While it could generate counterfactuals that changed the predicted outcome, some of the resulting trajectories showed abrupt changes that don’t align well with physiological expectations. This could reduce trust in real-world settings where clinicians rely on coherent time series patterns.

The neural models GRU and N-BEATS were more adaptive and expressive. GRUs, with their ability to model time dependencies and non-linear relationships, produced counterfactuals that stayed faithful to the original sequence while making smart, targeted adjustments. N-BEATS, with its deep architecture and built-in trend/seasonality modelling, generated smooth and well-aligned trajectories that captured both short- and long-term patterns effectively. This made it especially suitable for variable-length and dynamic clinical sequences.

One important strength across all models was their ability to make selective changes. Features like BMI, Creatinine, and Platelet Count, that are often less immediately tied to short-term survival, were mostly left untouched. Instead, the models focused their adjustments on more relevant features like Heart Rate, Troponin T, and Bicarbonate. This targeted behaviour adds to the interpretability and clinical credibility of the generated counterfactuals.

Altogether, the results point to real potential for using counterfactual modelling in clinical forecasting. These models don’t just help explain what the model saw, but they can also be used to simulate “what-if” scenarios, which could be useful for decision support, treatment planning, or even early warning systems in critical care.

Still, there are limitations to address. Our evaluation so far has been mainly qualita-

tive, and future work should include more rigorous quantitative metrics for plausibility, safety, and causal validity. Clinical deployment would also require expert review and integration with medical knowledge. And while we assume that changes to exogenous inputs cause changes in outcomes, this assumption should be carefully examined, especially when working with observational data that may include confounders.

In summary, our findings show that counterfactual generation in time series is both feasible and valuable. By comparing a variety of modelling strategies, we offer insights into how different models handle the trade-offs between flexibility, coherence, and interpretability, thus laying the groundwork for more transparent and clinically grounded machine learning in healthcare.

The evaluation of counterfactual generation reveals a nuanced trade-off between producing clinically plausible recommendations and maintaining fidelity to original patient data. Notably, the features exhibiting the largest average Euclidean distances, bolus insulin and carbohydrates, are consistent across both the general counterfactuals and the healthy patient comparison. This suggests that modifications in these features are central to achieving normoglycemic predictions, likely due to their direct influence on blood glucose regulation. However, these changes also represent substantial interventions that may be less feasible in real-world settings without clinical oversight.

Normalized Euclidean distances offer important insights into relative perturbations. Despite their smaller raw values, features such as exercise intensity exhibit meaningful normalized shifts, indicating that even minor changes in low-variance features can represent significant deviations. This highlights the importance of scale-aware evaluation when interpreting counterfactual modifications.

The comparison with healthy patient samples further validates the biological plausibility of the generated counterfactuals. The relatively small target Euclidean distances (ranging from 1.72 to 2.42) suggest that the model is capable of producing healthy-like recommendations with minimal deviation. Nevertheless, variability across normalized distances per feature and per pair illustrates that individualized pathways to health vary and may require adaptive adjustment strategies.

Global evaluation metrics paint a broader picture of the counterfactual landscape. The extremely high fraction of changed values (99.8%) indicates a pervasive transformation across input dimensions, aligning with the large average z-score change of nearly 38. While such changes drive effective prediction shifts, they also reflect a high degree of intervention intensity, which could pose challenges for clinical feasibility.

Encouragingly, the local outlier factor (LOF) score of -1.0 suggests that the counterfactuals remain within the natural data manifold, thereby enhancing their interpretability and realism. However, only 41.7% of counterfactuals fall within the predefined bounds, indicating that further refinement is needed to align generated samples with strict clinical or physiological constraints. The violation RMSE provides a quantitative measure of these infractions, helping to identify areas for constraint optimization.

Finally, similarity metrics reveal promising alignment with healthy baselines. A cosine similarity of 0.91 demonstrates that the counterfactual directions are well-aligned with healthy data trajectories. Meanwhile, the mean absolute difference of 6.87 provides a moderate benchmark of absolute deviation, suggesting a reasonable compromise between health alignment and intervention cost.

In summary, the results highlight the method’s strength in generating effective, realistic, and health-aligned counterfactuals, while also identifying opportunities to enhance con-

straint satisfaction and minimize excessive perturbations. These insights are crucial for translating model recommendations into clinically actionable and safe interventions.

5.3 Limitations and Future Work

While our evaluation framework provides valuable insights into the behaviour and plausibility of generated counterfactual interventions, there are several important limitations to note. First and foremost, our research team does not include healthcare professionals, which constrains our ability to fully assess the clinical relevance and real-world feasibility of the proposed adjustments. Although we rely on publicly available physiological bounds and established metrics, only domain experts can judge whether the magnitude, timing, and combination of interventions (e.g., changes in insulin dosing or carbohydrate intake) would be safe, practical, and ethically acceptable in actual diabetes management.

Another limitation is that our counterfactual generation is guided primarily by mathematical objectives, such as euclidean distances, z-scores, and manifold inlier scores, rather than by patient-centric considerations such as behavioural adherence, lifestyle constraints, or individual variability in insulin sensitivity. Consequently, the suggested interventions may overlook factors like meal timing, stress, or concurrent medications, which can strongly influence blood glucose dynamics.

From a methodological standpoint, our current approach treats each feature independently when computing distances and normalizing by global variability. In practice, exogenous inputs interact in complex, non-linear ways; future work should explore joint or conditional distance measures and incorporate domain-informed regularization to capture these dependencies more faithfully.

To address these limitations, we propose two key directions for future work. First, collaboration with healthcare professionals, including endocrinologists, diabetes educators, and clinical researchers, will be essential to validate the clinical appropriateness of our counterfactuals. Such collaboration can help refine the constraint sets, determine realistic intervention magnitudes, and guide the design of pilot studies to assess patient acceptance and safety in controlled environments.

Second, future work should aim to incorporate a richer clinical context and greater personalization. By integrating additional patient metadata such as age, weight, insulin sensitivity profiles, and lifestyle habits, the model can generate more individualized and clinically meaningful counterfactuals. Moreover, embedding behavioral models that estimate adherence likelihood, along with multi-objective optimization techniques, may help balance physiological targets with patient preferences and quality-of-life considerations.

By combining rigorous quantitative evaluation with deep domain expertise and patient-centric modelling, future iterations of this work can move toward practical, clinically actionable decision support tools for personalized diabetes management.

6 Conclusion

This thesis presents a novel framework for counterfactual time series forecasting that focuses on modifying exogenous variables within the forecast horizon to achieve desired outcomes, rather than altering historical data. This approach fills a critical gap in current research, offering a more actionable and interpretable alternative for clinical decision

support. By learning the relationship between forecasted targets and exogenous inputs using models such as SARIMAX, OLS, GRU, and N-BEATS, we enable the generation of realistic, constrained counterfactuals that respect the temporal dynamics of healthcare data.

Experimental results across three healthcare applications, blood glucose forecasting with OhioT1DM and SimGlucose datasets, and mortality prediction in HFpEF patients using MIMIC-IV, demonstrate the practical utility and robustness of the method. In multivariate forecasting, N-BEATS consistently outperformed GRU in both accuracy and temporal alignment, particularly for longer input sequences, making it the most suitable engine for our counterfactual pipeline. The counterfactual sequences produced using this model were not only effective in shifting predicted outcomes but also remained physiologically plausible and aligned with known clinical patterns.

Importantly, the analysis revealed that generating meaningful interventions often involves coordinated adjustments across multiple variables, mirroring real-world clinical strategies. However, limitations such as poor minority class detection in imbalanced datasets and the absence of clinical domain input highlight the need for further refinement. Future work should involve collaboration with healthcare professionals, incorporation of richer patient context, and optimization techniques that account for adherence, safety, and ethical considerations.

In summary, this work demonstrates that counterfactual generation for time series forecasting is both feasible and clinically relevant. It provides a foundation for building transparent, adaptive, and patient-centred decision support systems capable of suggesting personalized, data-driven interventions grounded in realistic future scenarios.

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A Appendix

A.1 Detailed Classification Metrics for MIMIC Dataset

The following tables provide comprehensive classification metrics (precision, recall, F1-score, and overall accuracy) for GRU and N-BEATS models across all patient clusters and gender subgroups in the MIMIC dataset.

Table 19: Classification Metrics for GRU and N-BEATS Models Cluster 0

Model	Class	Precision	Recall	F1-Score	
GRU	30 days	Survived	0.91	0.86	0.88
		Died	0.00	0.00	0.00
		Overall Accuracy		0.7882	
	1 year	Survived	0.90	0.91	0.90
		Died	0.00	0.00	0.00
		Overall Accuracy		0.8235	
N-BEATS	30 days	Survived	0.92	0.97	0.94
		Died	0.00	0.00	0.00
		Overall Accuracy		0.8922	
	1 year	Survived	0.90	0.96	0.93
		Died	0.06	0.02	0.03
		Overall Accuracy		0.8706	

Table 20: Classification Metrics for GRU and N-BEATS Models Cluster 1

Model	Class	Precision	Recall	F1-Score	
GRU	30 days	Survived	0.80	0.22	0.35
		Died	0.07	0.50	0.12
		Overall Accuracy		0.2486	
	1 year	Survived	1.00	0.05	0.10
		Died	0.10	1.00	0.19
		Overall Accuracy		0.1475	
N-BEATS	30 days	Survived	0.90	0.93	0.91
		Died	0.00	0.00	0.00
		Overall Accuracy		0.8424	
	1 year	Survived	0.91	0.97	0.94
		Died	0.26	0.08	0.13
		Overall Accuracy		0.8871	

Table 21: Classification Metrics for GRU and N-BEATS Models Cluster 2

Model	Class	Precision	Recall	F1-Score	
GRU	30 days	Survived	0.90	0.53	0.67
		Died	0.12	0.50	0.19
		Overall Accuracy		0.5291	
	1 year	Survived	0.91	0.61	0.73
		Died	0.14	0.50	0.21
		Overall Accuracy		0.5979	
N-BEATS	30 days	Survived	0.89	0.99	0.94
		Died	0.18	0.02	0.04
		Overall Accuracy		0.8807	
	1 year	Survived	0.90	0.88	0.89
		Died	0.15	0.18	0.16
		Overall Accuracy		0.7982	

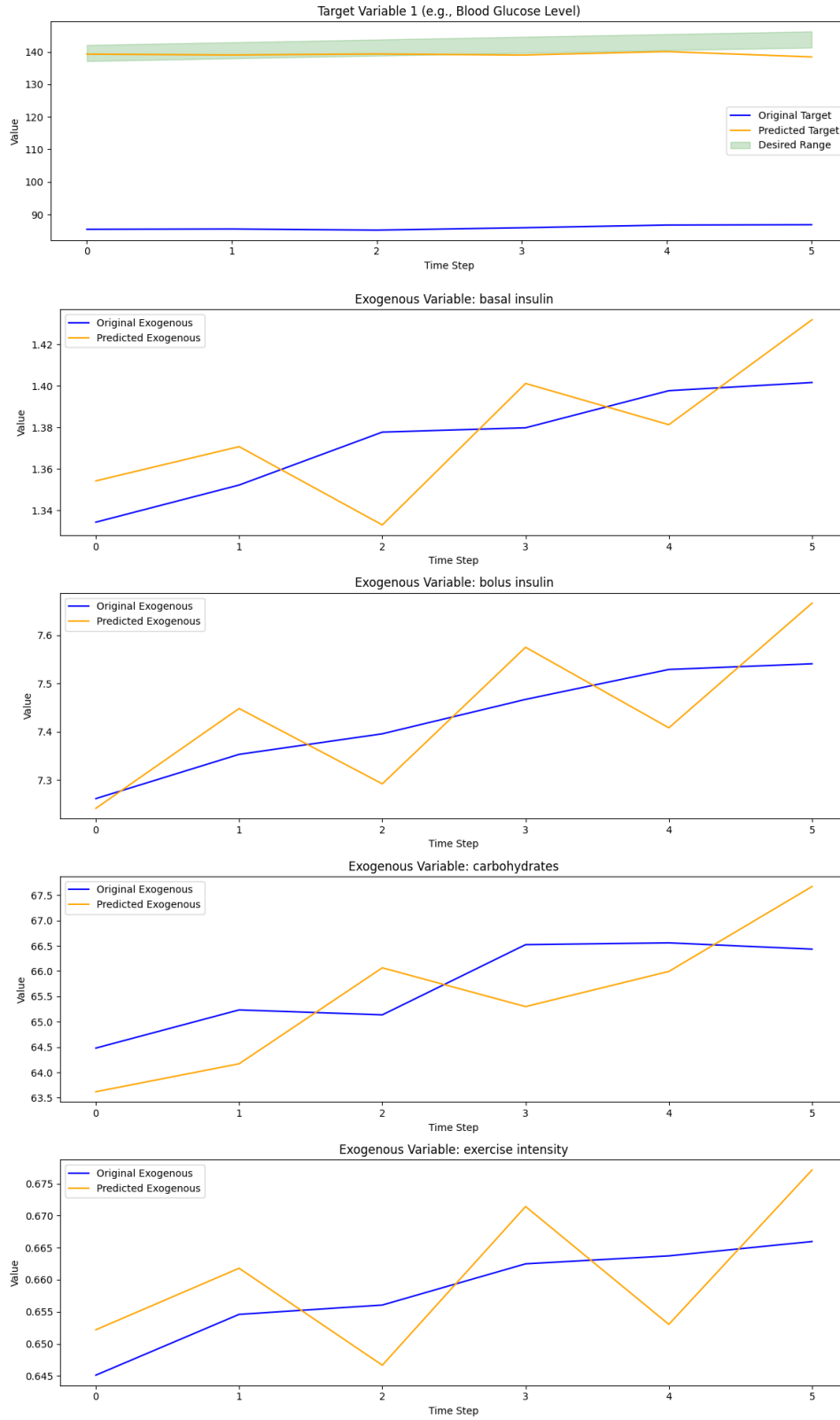


Figure 10: Example of counterfactuals generated using GRU for the OhioT1DM dataset. Top: Blood glucose levels for the original and counterfactual samples. Bottom: Comparison of exogenous variables (original in blue, counterfactual in yellow, bounds in green).

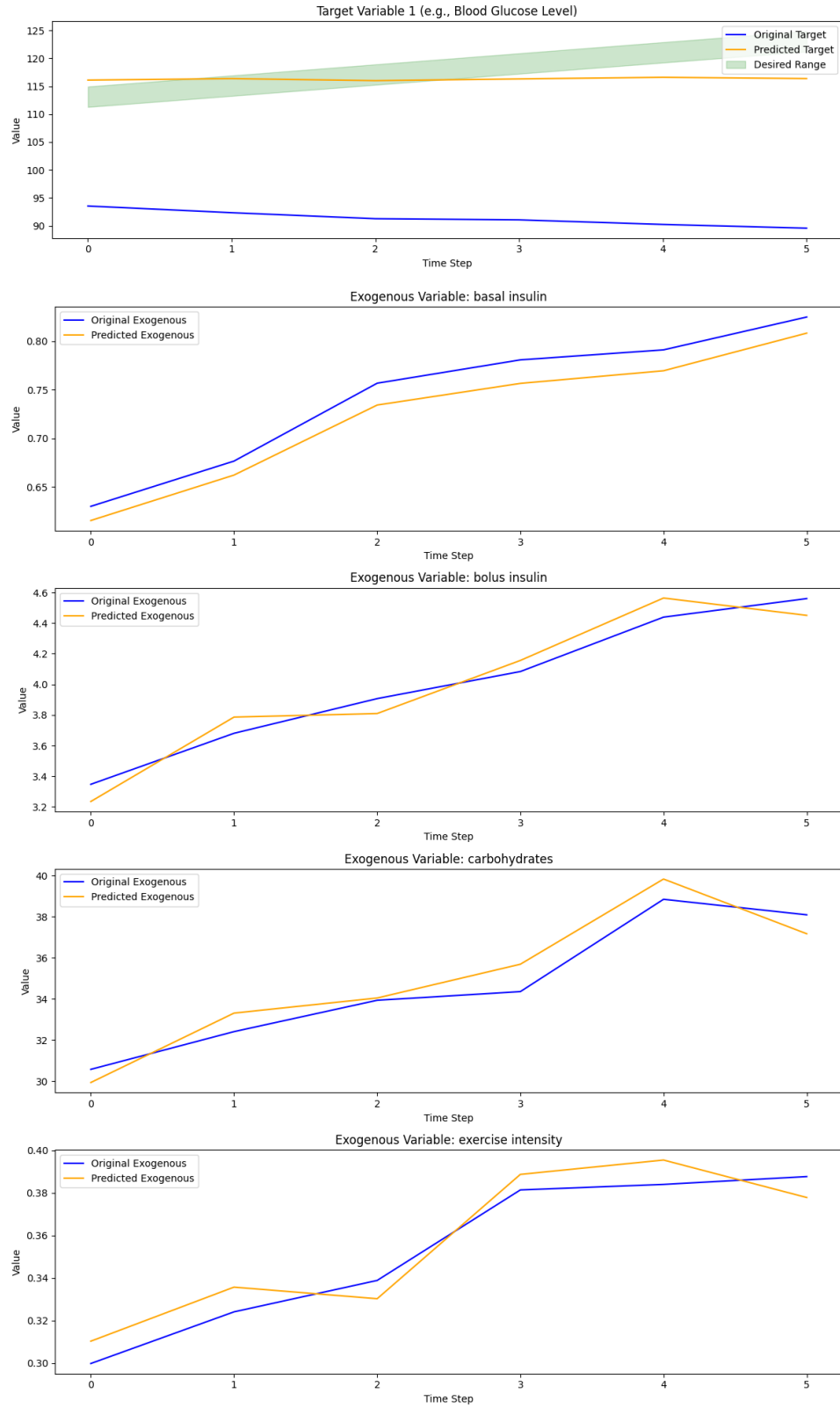


Figure 11: Example of counterfactuals generated using GRU for the OhioT1DM dataset. Top: Blood glucose levels for the original and counterfactual samples. Bottom: Comparison of exogenous variables (original in blue, counterfactual in yellow, bounds in green).

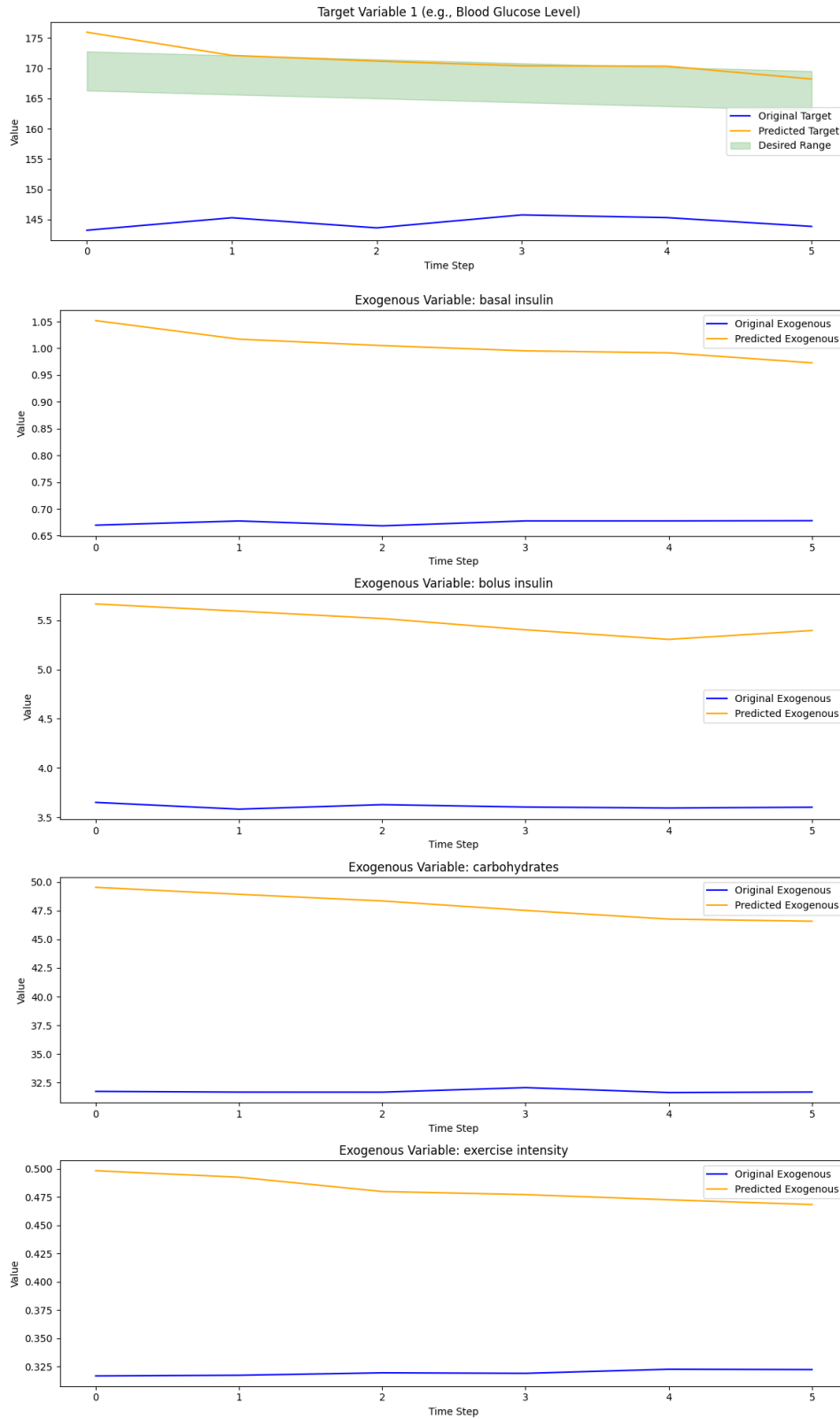


Figure 12: Example of counterfactuals generated using SARIMAX for the OhioT1DM dataset. Top: Blood glucose levels for the original and counterfactual samples. Bottom: Comparison of exogenous variables (original in blue, counterfactual in yellow, bounds in green).

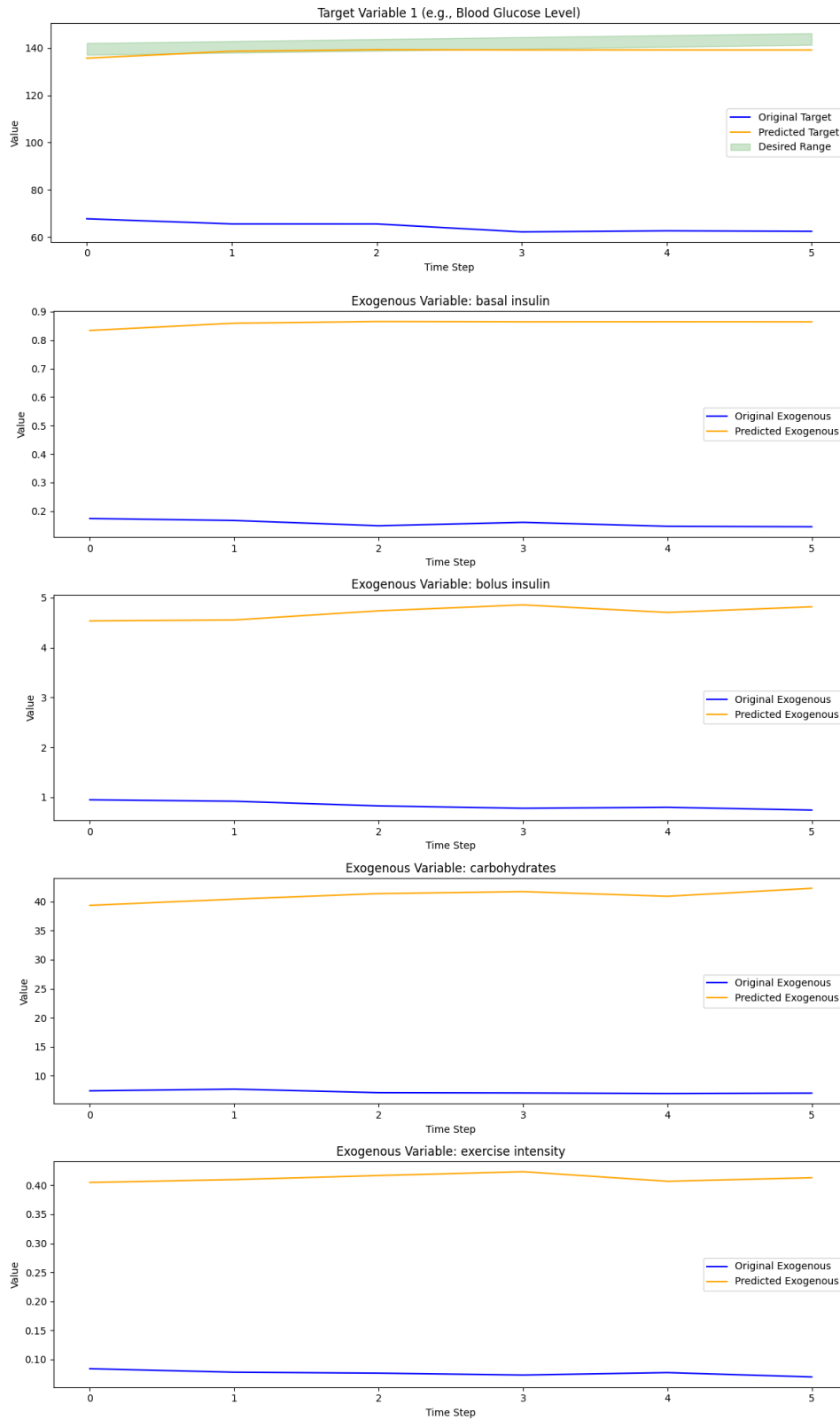


Figure 13: Example of counterfactuals generated using OLS for the OhioT1DM dataset. Top: Blood glucose levels for the original and counterfactual samples. Bottom: Comparison of exogenous variables (original in blue, counterfactual in yellow, bounds in green).

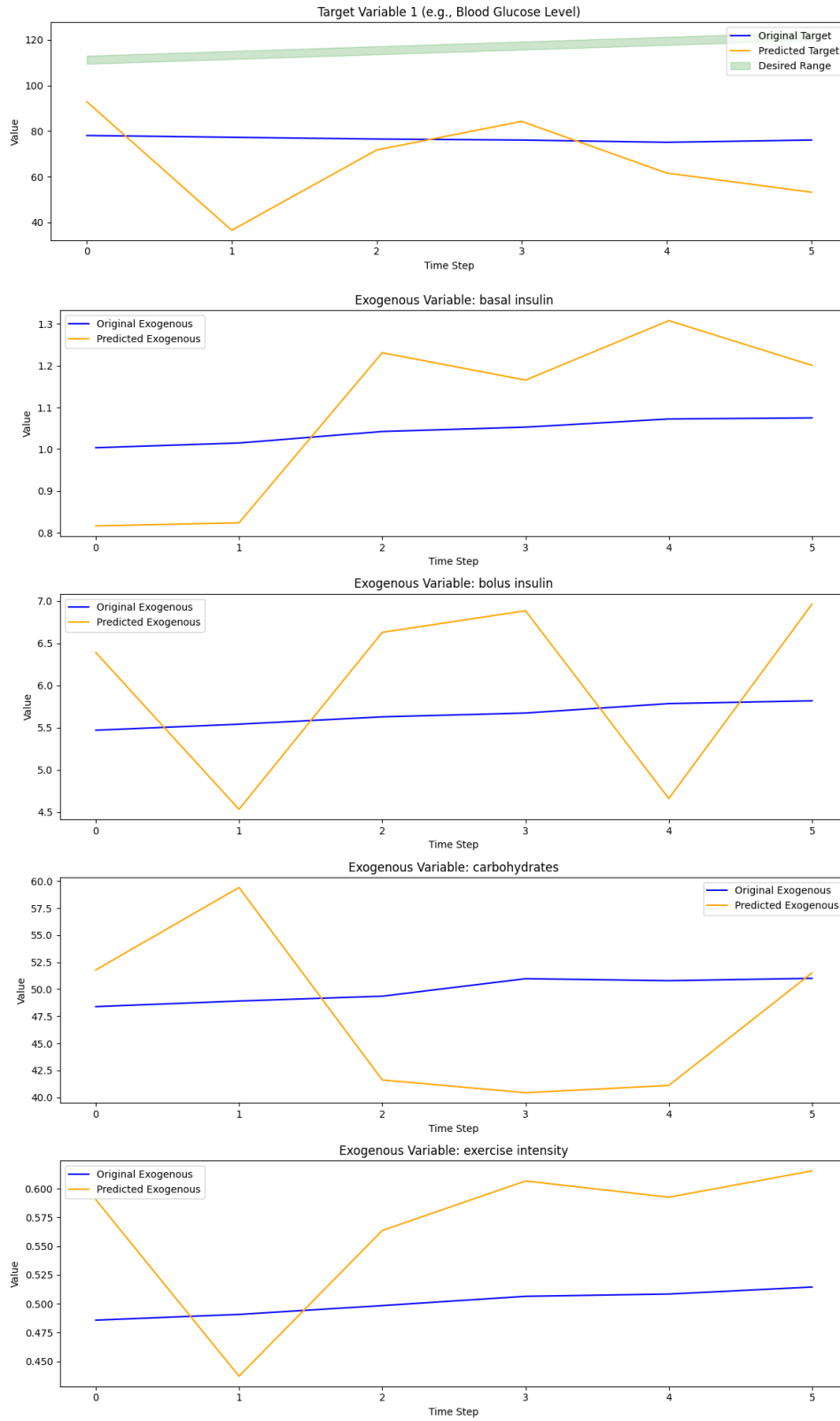


Figure 14: Example of counterfactuals generated using N-BEATSx for the OhioT1DM dataset. Top: Blood glucose levels for the original and counterfactual samples. Bottom: Comparison of exogenous variables (original in blue, counterfactual in yellow, bounds in green).

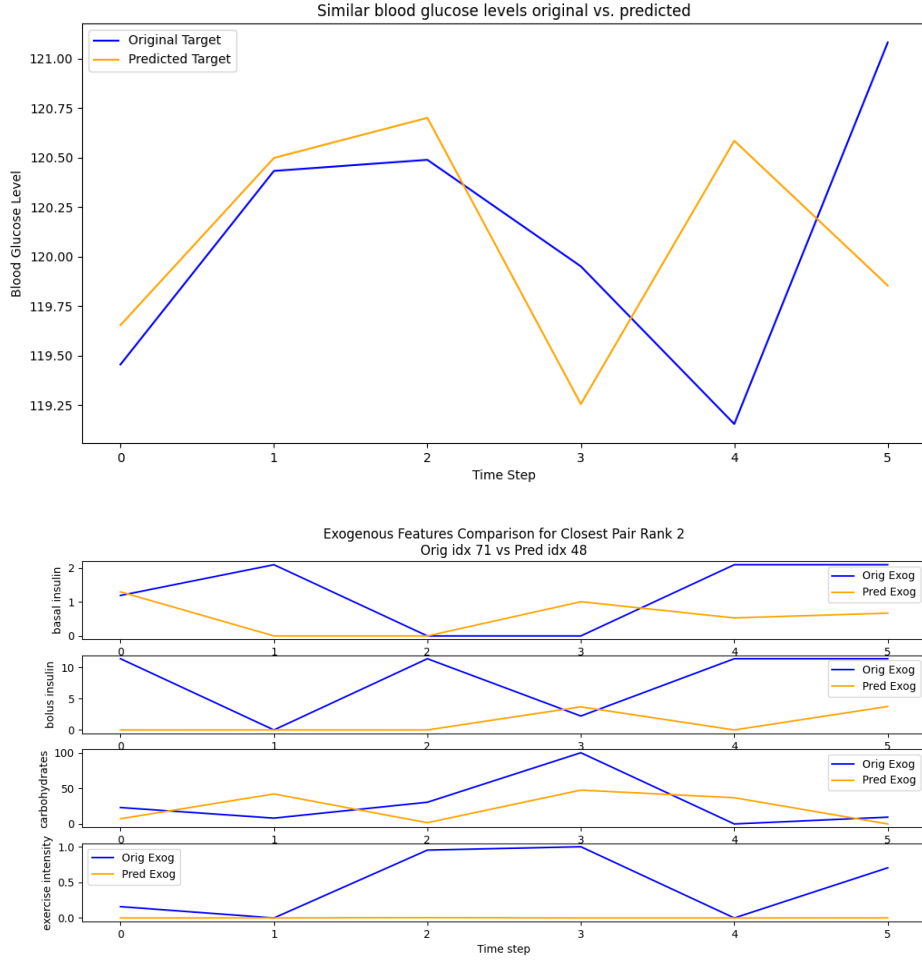


Figure 15: Comparison of two samples from the OhioT1DM dataset with similar blood glucose levels. Top: Blood glucose targets. Bottom: Exogenous variables for both samples (original in blue, comparison sample in yellow).

Table 22: Classification Metrics for GRU and N-BEATS Models Cluster 3

Model	Class	Precision	Recall	F1-Score	
GRU	30 days	Survived	0.95	1.00	0.97
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9487	
	1 year	Survived	0.95	1.00	0.97
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9487	
N-BEATS	30 days	Survived	0.95	1.00	0.97
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9484	
	1 year	Survived	0.95	1.00	0.97
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9487	

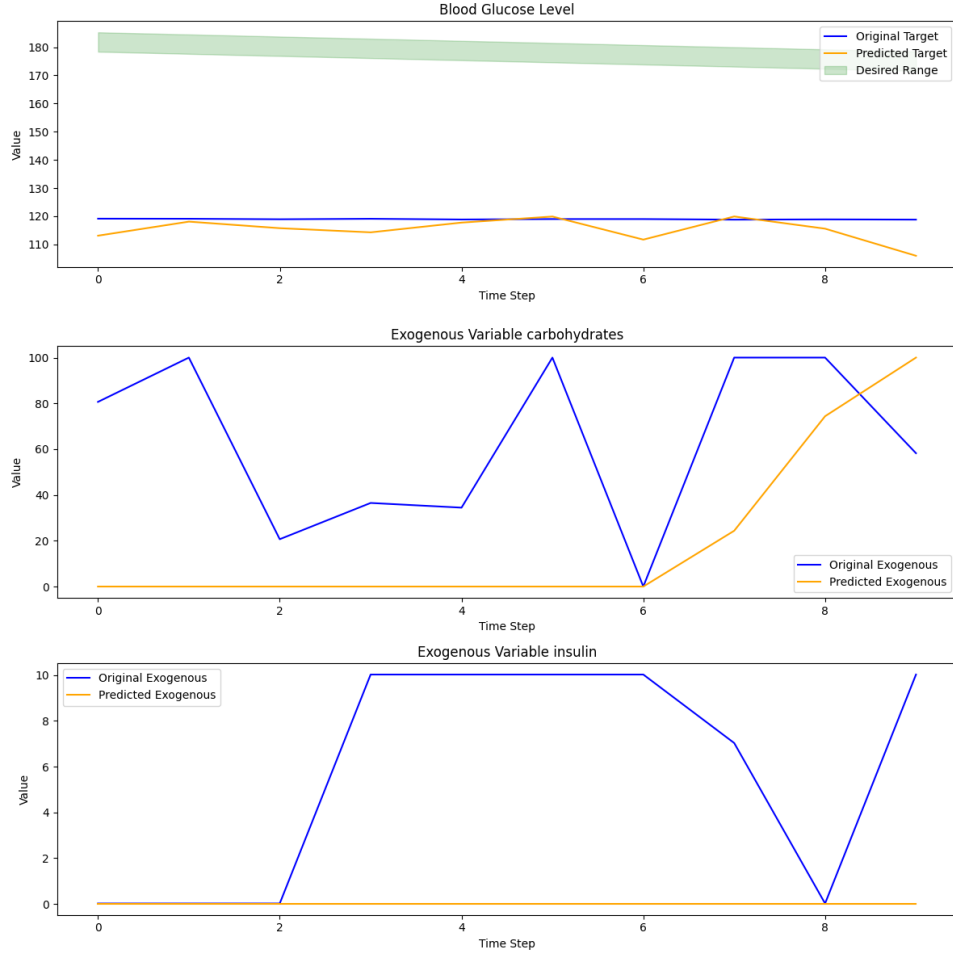


Figure 16: Example of the counterfactuals generated for the SimGlucose dataset, with the blood glucose level within the desired bounds. The different exogenous variables, with the original values in blue and the predicted in yellow.

Table 23: Classification Metrics for GRU and N-BEATS Models Female

Model	Class	Precision	Recall	F1-Score	
GRU	30 days	Survived	0.94	1.00	0.97
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9414	
	1 year	Survived	0.94	1.00	0.97
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9369	
N-BEATS	30 days	Survived	0.94	1.00	0.97
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9398	
	1 year	Survived	0.94	1.00	0.97
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9362	



Figure 17: Example of the counterfactuals generated for the MIMIC dataset using SARI-MAX, with survival. The different exogenous variables, with the original values in blue and the predicted in yellow.



Figure 18: Example of the counterfactuals generated for the MIMIC dataset using OLS, with survival. The different exogenous variables, with the original values in blue and the predicted in yellow.



Figure 19: Example of the counterfactuals generated for the MIMIC dataset using GRU, with survival. The different exogenous variables, with the original values in blue and the predicted in yellow.

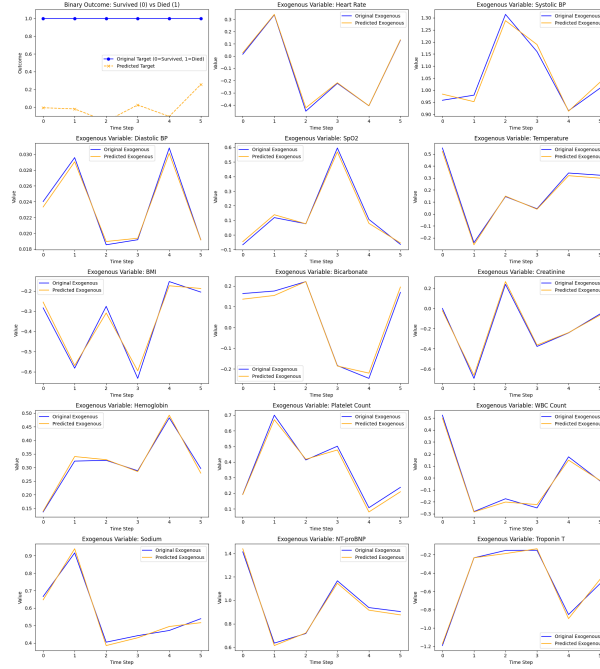


Figure 20: Example of the counterfactuals generated for the MIMIC dataset using NBEATS, with survival. The different exogenous variables, with the original values in blue and the predicted in yellow.

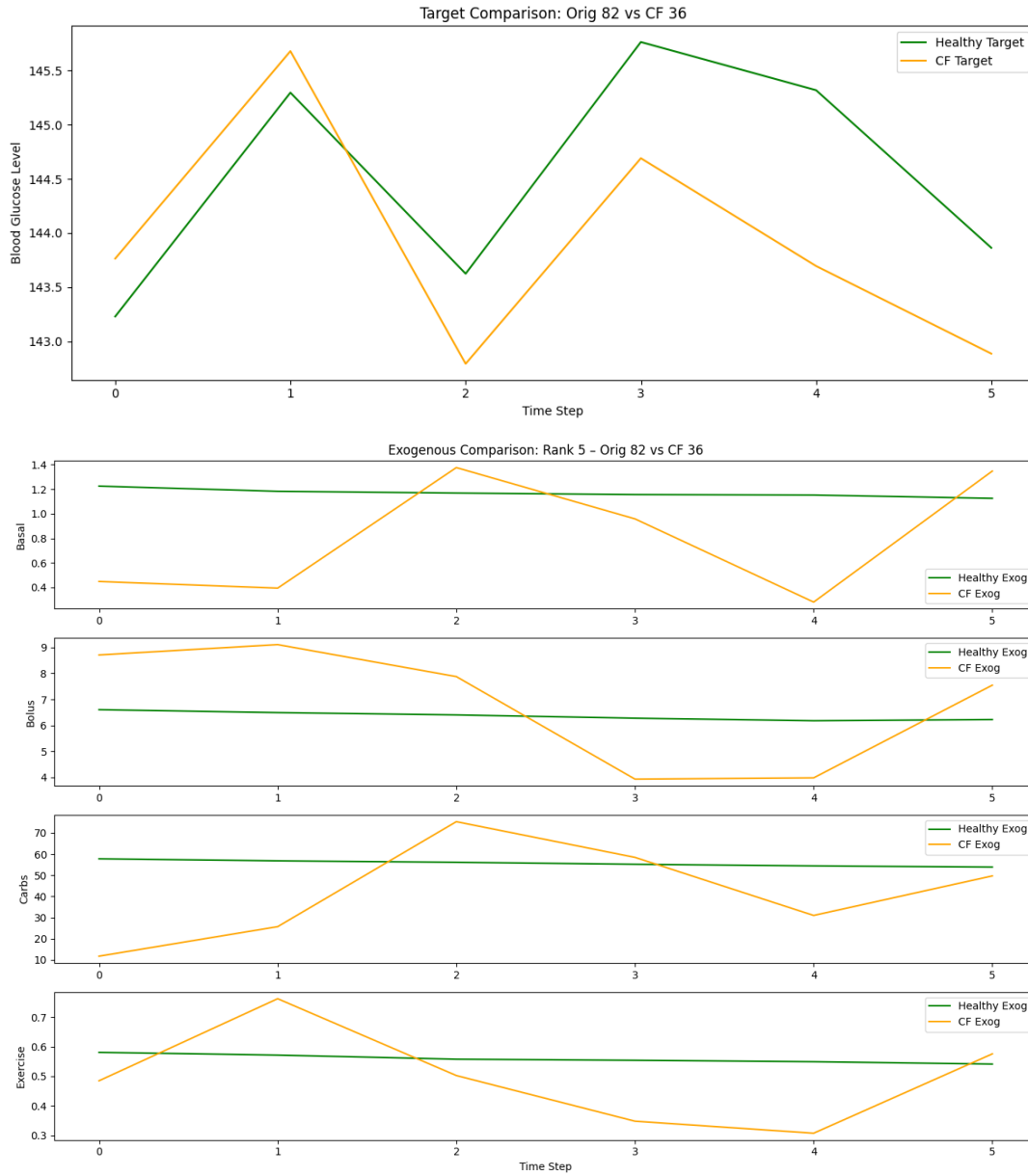


Figure 21: Example of the comparison between generated counterfactuals and similar original data for the OhioT1DM dataset. The target and the different exogenous variables, with the original values in green and the predicted in yellow.

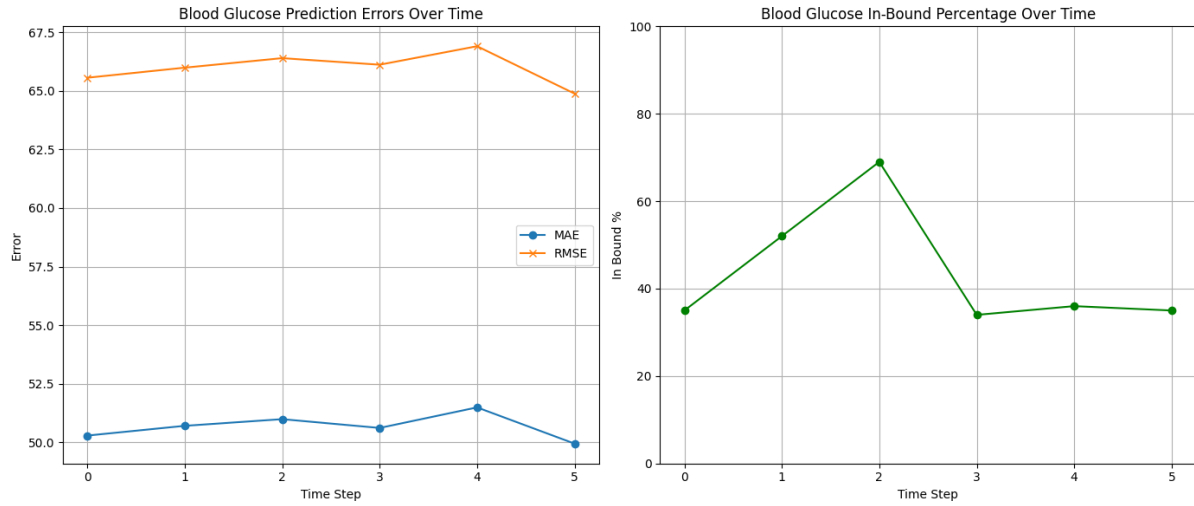
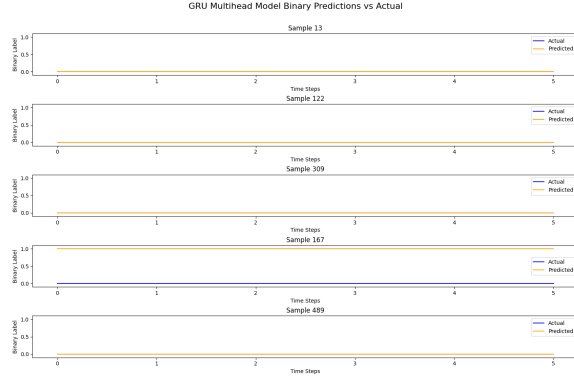


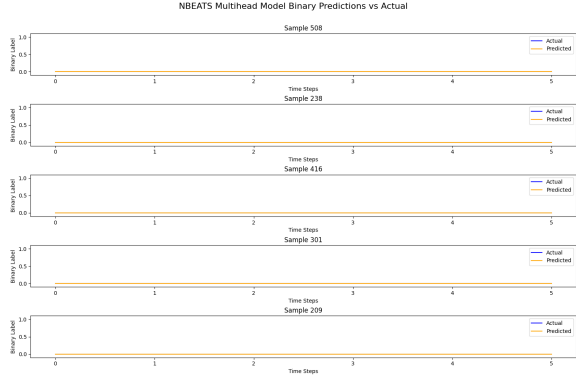
Figure 22: Prediction Errors and In-Bound percentage over time.

Table 24: Classification Metrics for GRU and N-BEATS Models Male

Model	Class	Precision	Recall	F1-Score	
GRU	30 days	Survived	0.93	1.00	0.96
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9299	
	1 year	Survived	0.93	1.00	0.96
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9299	
N-BEATS	30 days	Survived	0.93	1.00	0.96
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9298	
	1 year	Survived	0.93	1.00	0.96
		Died	0.00	0.00	0.00
		Overall Accuracy		0.9296	

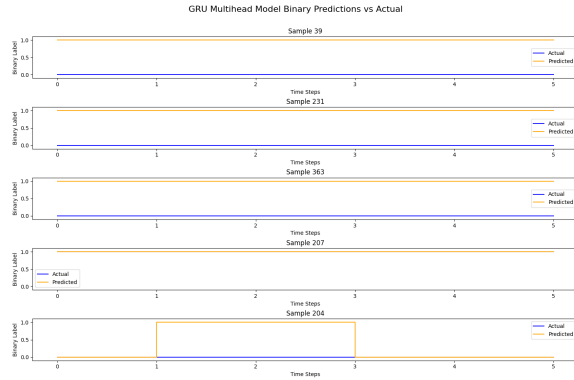


(a) Results of the multivariate forecasting using GRU.

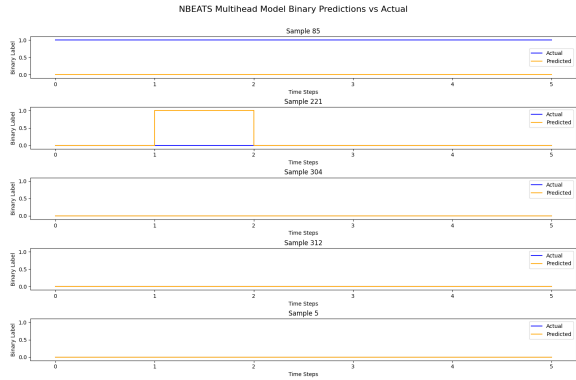


(b) Results of the multivariate forecasting using NBEATS.

Figure 23: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

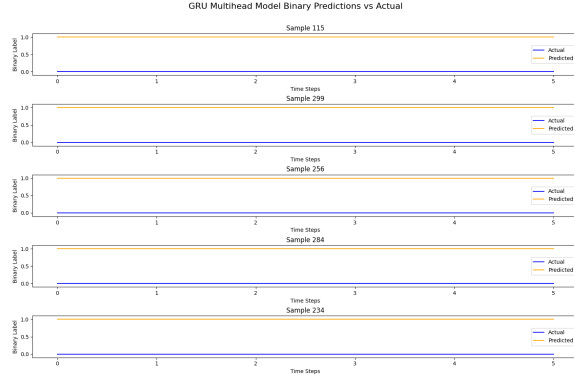


(a) Results of the multivariate forecasting using GRU.

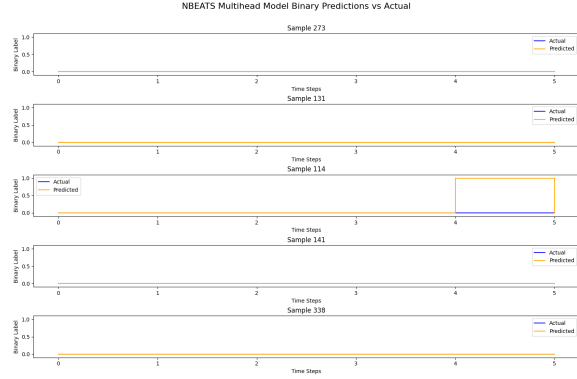


(b) Results of the multivariate forecasting using NBEATS.

Figure 24: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

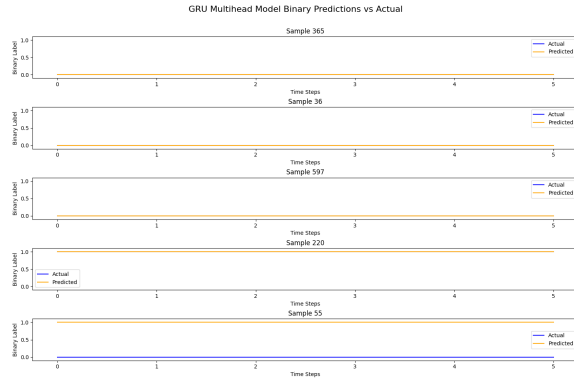


(a) Results of the multivariate forecasting using GRU.

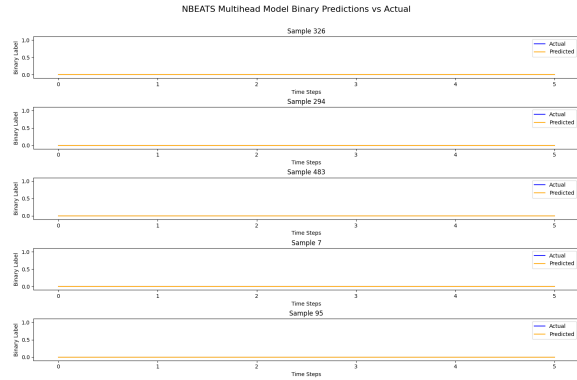


(b) Results of the multivariate forecasting using NBEATS.

Figure 25: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

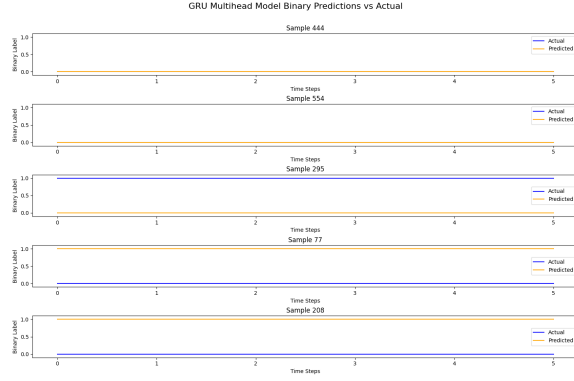


(a) Results of the multivariate forecasting using GRU.

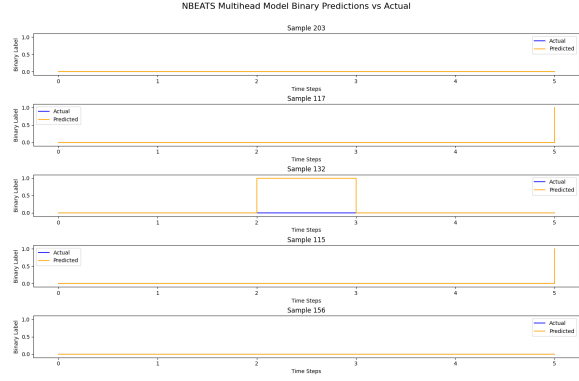


(b) Results of the multivariate forecasting using NBEATS.

Figure 26: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

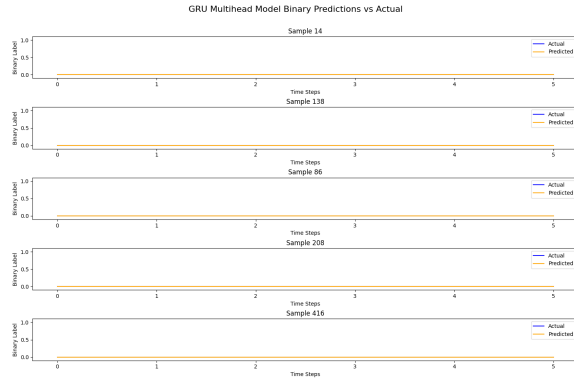


(a) Results of the multivariate forecasting using GRU.

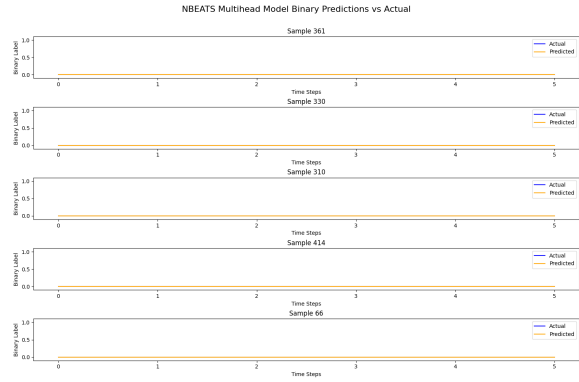


(b) Results of the multivariate forecasting using NBEATS.

Figure 27: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

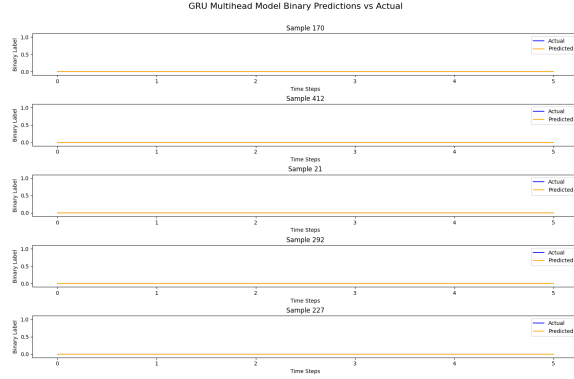


(a) Results of the multivariate forecasting using GRU.

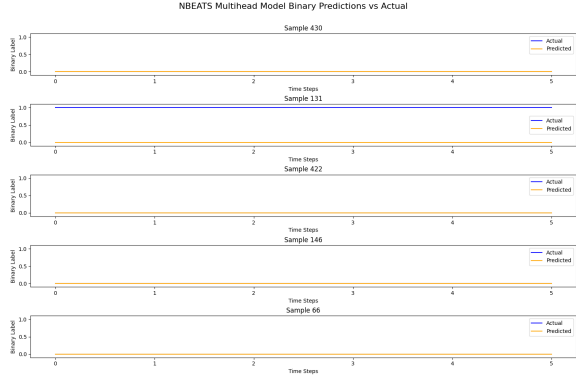


(b) Results of the multivariate forecasting using NBEATS.

Figure 28: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

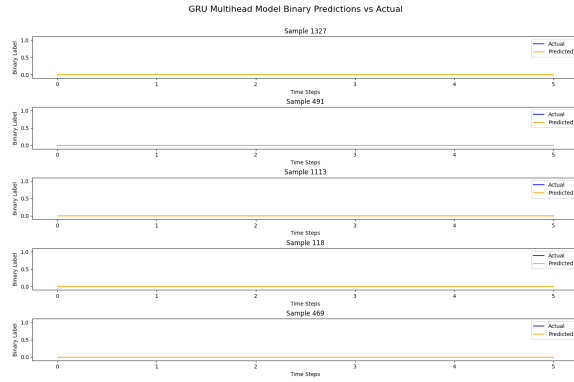


(a) Results of the multivariate forecasting using GRU.

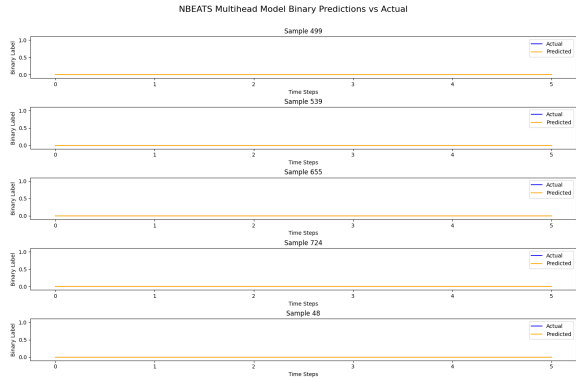


(b) Results of the multivariate forecasting using NBEATS.

Figure 29: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

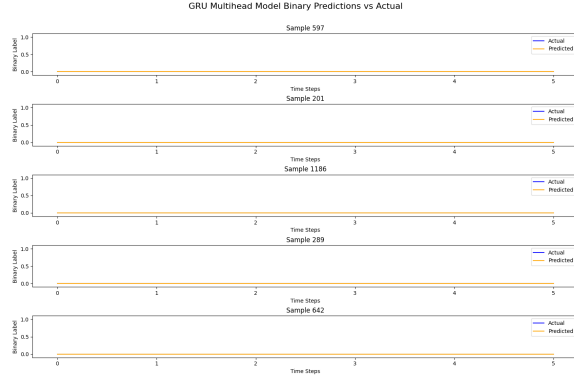


(a) Results of the multivariate forecasting using GRU.

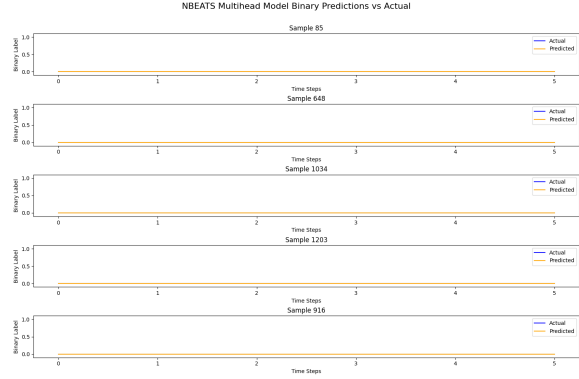


(b) Results of the multivariate forecasting using NBEATS.

Figure 30: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

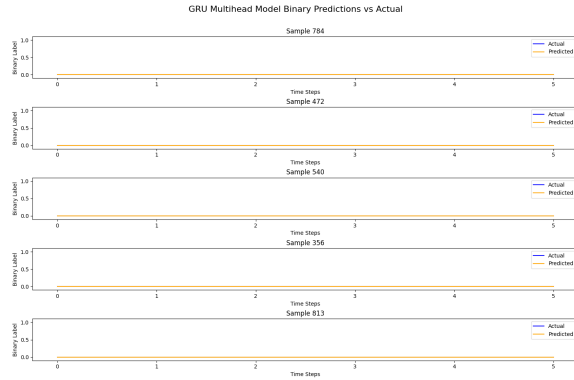


(a) Results of the multivariate forecasting using GRU.

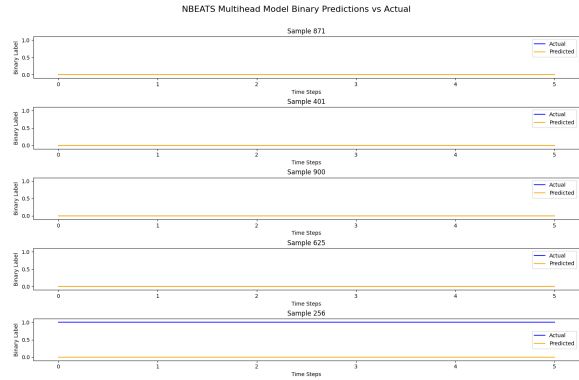


(b) Results of the multivariate forecasting using NBEATS.

Figure 31: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.

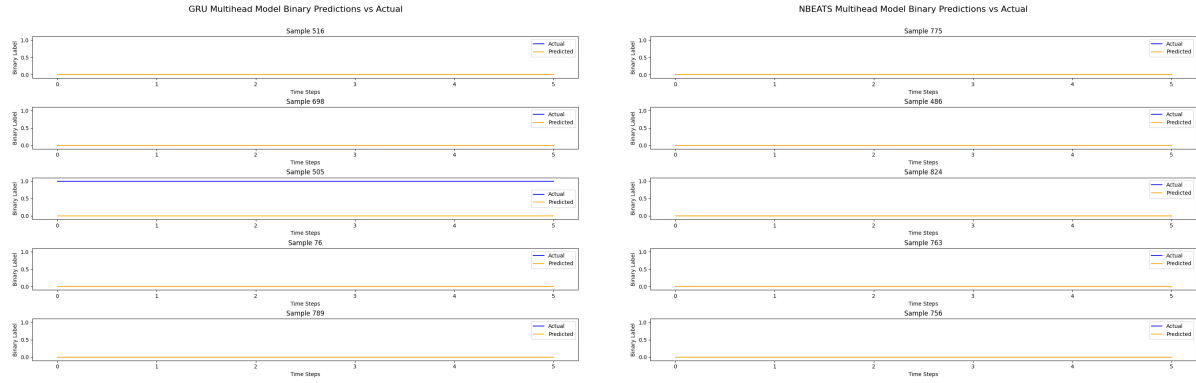


(a) Results of the multivariate forecasting using GRU.



(b) Results of the multivariate forecasting using NBEATS.

Figure 32: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.



(a) Results of the multivariate forecasting using GRU. (b) Results of the multivariate forecasting using NBEATS.

Figure 33: Results of the multivariate forecasting for the MIMIC dataset with 30-day mortality as target variable, back horizon = 12, and forecast horizon = 6, showing the accuracy of the forecasting.